THE

SCARA Robot

SCARA Robot THE600-T-E Ceiling Mount Specifications Manual

Rev. 2

November 30, 2022

Important

- Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
- Before operating the industrial robot, read through and completely understand this manual.
- After reading through this manual, keep it nearby for future reference.

Original Instructions

Shibaura Machine

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Introduction

Thank you very much for your purchase of SHIBAURA MACHINE's robot system.
This manual describes the basic specifications of the industrial robot, and how to unpack and install the robot, how to connect wiring, how to install the hand, and how to maintain and inspect the robot.
This manual omits content that is identical to the model with standard specifications, and so use this manual together with the instruction manual “STA1211: THE600-E Robot Manual”.
Before actually unpacking the machine, be sure to look through the manual once.
Before using this manual, read “STEA1212: THE600-E Safety Manual” to ensure that you fully understand the implemented safety measures.
This manual has the following chapters.

Specifications

Chapter 1: Specifications
This chapter describes the basic specifications and names of the robot parts.

Chapter 2: Functions
This chapter describes the key points for using the functions provided with the robot.

Installation

Chapter 3: Transportation
This chapter describes how to transport the robot in its packed state.

Chapter 4: Unpacking
This chapter describes how to unpack the robot.
It also describes the key points for temporarily storing the unpacked robot.

Chapter 5: Transfer
This chapter describes how to move the unpacked robot to the installation site.
It also provides the external dimensions and weight of the robot.

Chapter 6: Installation of the Robot
This chapter describes the installation environment, required space, and installation procedure for the robot.

Chapter 7: Hand and Other Settings
This chapter describes how to mount the hand to the robot.
Finally, this chapter provides the allowable payloads for the hand.

Chapter 8: User Wiring
This chapter describes how to route the wiring to the hand.

Chapter 9: Teaching
This chapter describes the key points when teaching the robot.
This also describes how to guide the robot manually.

Chapter 10: Operation
This chapter describes how to rescue someone who is trapped by the robot and operation when the robot is brought to an emergency stop.
It also describes the key points for preparing to repack an installed robot.

Chapter 11: Repacking
This chapter describes the key points for repacking an installed robot.
It also describes the key points for transporting a repackaged robot.

Maintenance Manual
Chapters 12 and 13: Maintenance
These chapters describe the maintenance procedures for the various robot sections.

Chapter 14: Origin Position Setting and Position Detector Error
This chapter describes the origin position setting procedure for the robot and the corrective action taken if a position detector error occurs.

Chapter 15: Service Parts
This chapter describes service replacement parts of the robot.

Chapter 16: Disposal
This chapter describes disposal of the robot and its parts.

Manufacturer

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If you have purchased this robot outside of Japan, please contact the company that you purchased it from.

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Use Range

This product is the horizontally articulated type industrial robot in which the manipulator has two horizontally rotating joints and the mechanical interface on its tip moves vertically and rotates horizontally.
The product has been developed with intention of adapting it to automation of wide range of work including conveying work such as movement and alignment of workpieces and assembly work such as insertion of parts and screw tightening.
The customer should also use this product with the aim to adapt it to automation of these work and incorporate it to the automation equipment.

This product is not intended to apply it to work that contacts a human (such as work related to medical care and nursing care), incorporate it to the equipment that supports human life (such as life support device and incubator), incorporate it to the equipment on which a human rides (including transport equipment such as car, aerospace equipment such as airplane, and game equipment such as roller coaster), and incorporate it to the equipment that protects human life and human body (such as safety device).
This product is not intended to incorporate it to the equipment aimed to control large quantity of heat (such as nuclear power controller and incinerator controller).
This product is prohibited from being used to apply it to work or incorporate it to the equipment that is likely to threaten human life and harm a human body directly due to a failure or malfunction. If this product is used, it will be guaranteed under customer’s responsibility.

Use Method

The use conditions for the THE600-T-E robot are described. If the product is used in the state that differs from the conditions, unexpected risk may occur. Abide by the conditions for use.

1. Main Specifications of Product: See Specifications "1.4 Specifications Table" and Installation "6.1 Installation Environment."

2. Restrictions on Use: Table 1 and Table 2 give "Restricted Specification Sheet" indicating restrictions on use.

Table 1. Restricted Specifications Sheet

1. Purpose and application for using product (usage restrictions)	
1) Intended use	To make the robot perform operations for specific purposes (item transport, assembly, inspection, machining, etc.) (1) Transportation [1] Obtain a transportation route to the unpacking site. [2] Use a forklift or other equipment to lift the packing box. [3] Move the packing box to the unpacking site. (2) Unpacking [1] Obtain the space required for unpacking. [2] Cut the packing straps and unpack. [3] Cut the packing straps securing the robot. [4] Peel off the tape from the controller packing box, and unpack. [5] Check that the robot and controller are not damaged. [6] Check the accessories and their quantities. - When transferring the robot to the installation site [7] Dispose of, or store and manage the packing materials. - When storing the robot [7] Secure in place in the same way as the state at delivery, and repack. [8] Move the packing box to the storage site. (3) Transfer - Robot [1] Check the orientation of the robot during transfer. [2] Check that the robot fixing jigs are mounted. [3] If using a crane, mount the supplied eyebolts. [4] Check the tightness of the robot fixing jig fixing bolts. [5] Use a crane or cart to transfer to the installation site. - Controller [1] Use a cart to transfer to the installation site.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
1) Intended use (continued)	(4) Installation [1] Use bolts to secure the robot to the frame and other parts. [2] Fabricate any needed cables. [3] Remove the robot fixing jigs and eyebolts. [4] Install the controller and connect it to the robot. [5] Push the robot by hand to move to a location where it is easy to perform work. [6] Mount the hand. [7] Connect the required external devices. (5) Teaching [1] Move the robot to the desired position or orientation by guiding it with a teach pendant or pushing it by hand. [2] Teach the position where the robot will operate. [3] Create the operation sequence (program, etc.). [4] Make sure that the signals and data are transferred to and from the external equipment. [5] Check operation at low speed (250 mm/sec or less). [6] While gradually increasing the operating speed, make sure that the operation matches the specific purpose. (6) Operation [1] Start robot operation. [2] Make sure that operation is being performed to match the specific purpose at the specified speed. [3] Stop robot operation. (7) Repacking [1] Remove the hand. [2] Move the robot to the position and orientation where the fixing jig can be mounted by guiding it with a teach pendant or pushing it by hand. [3] Mount the fixing jig. [4] Turn off the power, and remove the wiring. [5] Obtain the space needed for packing. [6] Use a crane to transfer the robot, and mount the supplied eyebolts. [7] Transfer the robot and controller to the packing site. [8] Pack the robot and controller so that they are in the original delivery state. [9] Obtain a transportation route to the transportation equipment (truck, etc.). [10] Use a forklift to lift the packing box. [11] Load the packing box onto the transportation equipment.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
1) Intended use (continued)	(8) Maintenance Work performed when the primary power is off: [1] Check the bolts for any looseness and backlash and re-tighten them. [2] Make sure that the cable clamp is not loose. [3] Check for any damage. [4] Clean off any dust. [5] Inspect and apply corrosion inhibitor. [6] Check for any wear on the cable and air tube. [7] Check that there is no misalignment between the motion of each joint axis and the origin position. [8] Replace the memory backup batteries in the controller. Work performed when the primary power is on: [1] Check the grease state and apply grease (when the servo power is off). [2] Check for any abnormal vibrations and sounds during operation (when the servo power is on). [3] Check the teaching point for any misalignment (when the servo power is on). [4] Replace the backup batteries for the robot position detector (when the servo power is off). (9) Disposal [1] Remove the hand. [2] Move the robot to the position and orientation where the fixing jig can be mounted by guiding it with a teach pendant or pushing it by hand. [3] Mount the fixing jig. [4] Turn off the power, and remove the wiring. [5] Use a crane to transfer the robot, and mount the supplied eyebolts. [6] Transfer the robot and controller to the disposal site. [7] Dispose of the robot using a method compliant with your country’s laws and regulations.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
2) Foreseeable misuse	- Use for work which applies excessive impact to the robot or to which excessive impact is applied by the robot - Modification of robot, controller, and optional parts, or supplying power to robot, controller, and optional parts while their covers are removed - Operation in other installation state than the specified (floor, suspended, and wall-mounted) - Operation on a frame or other structure that does not satisfy the specified strength - Operation under conditions not satisfying the specified installation requirements - Entry into the hazard zone of the robot by mistakenly assuming that operations would have stopped although the stop is just for waiting for an operation signal during operation - Usage under conditions where the specification values are exceeded or usage under improper settings - Entry into the hazard zone of the robot when the servo power is on - Stop of the robot caused by power off during operation or when the servo power is on - Unintended operation caused by inadvertently pressing of a button - Operating without checking operation in MANUAL mode - Operating without checking the tightness of the robot and hand set bolts - Unintended operation due to improper wiring - Unintended operation due to incorrect program selection - Riding on or leaning against the robot or hanging on the arm
3) Unexpected start	- Malfunction due to excessive noise - Robot start due to start signal unexpectedly sent from peripheral equipment to the robot - Unintended start due to abnormal communication data - Malfunction due to voltage fluctuation - Power on due to powering after power outage - Malfunction due to failure or other cause

Table 2. Restricted Specifications Sheet

2. Replacement or other modification of program components (time restrictions)	
1) Mechanical restrictions	- Before operating the robot, perform operation at a low speed (20% or less of maximum speed) for five (5) minutes - Replace consumable parts as required or periodically by conducting daily inspection, regular inspection (every 3 and 6 months and every year) and overhauls (every 5 years).
2) Electrical restrictions	

Table 3. Restricted Specifications Sheet

3 Movable range of product (restriction on space)		
1) Movable range	1st axis 2nd axis 3rd axis 4th axis Additional axis - -	Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with individual specifications. - -
2) Interface	Transportation: Unpacking: Transfer: Installation: Teaching: Operation: Repacking: Maintenance: Disposal: 	The packing box shall be able to be transferred by a transportation worker. The packing box shall be able to be carried in. Space of at least 2 m^3 shall be secured. The robot shall be able to be hoisted by a crane. The robot shall be able to be lifted by two or more people and loaded onto a cart or similar device. There shall be no obstacles on the route from the unpacking site to the installation site. The product shall be installed at a location suitable for the outline and work envelope of the robot. The product shall be installed at a location suitable for the outline and that provides adequate ventilation of the controller. The cable shall be installed at a location where it will not be stepped on by operators. Teaching work shall be able to be performed from outside the hazard zone of the robot. A fence or enclosure shall be installed to prevent any operator from approaching the hazard zone of the robot. The robot shall be able to be operated from outside the hazard zone of the robot. An operator in the fence or enclosure shall be visible from the operation location. Work shall be able to be performed to remove the hand, mount the fixing jig, separate the robot and controller, and remove the robot and controller from the frame. The robot shall be able to be moved from the installation site to the packing site. The repacking work shall be able to be performed. A forklift or other equipment shall be able to be used to lift the packing box. Space required for maintenance shall be obtained. After maintenance, operation shall be able to be checked from outside the hazard zone of the robot. Work shall be able to be performed to remove the hand, mount the fixing jig, separate the robot and controller, and remove the robot and controller from the frame. The robot shall be able to be moved from the installation site to the disposal location.

Table 3. Restricted Specifications Sheet

3. Movable range of product (restriction on space)	
3) Work environment	- A fence, enclosure, or other protection shall be provided that surrounds the hazard zone of the robot to prevent people from entering the zone during operation. - A display indicating the robot operation status shall be provided to indicate whether the robot is operating or stopped. - The drive power shall be turned off whenever a person enters the robot hazard zone. - Lighting shall be provided for enabling visual checks by operators. - To prevent operators from slipping, the floor shall not be covered with oil or similar substances. - No noises that impair attention shall be emitted.

Table 4. Restricted Specifications Sheet

4. Life cycle of product
In the product life cycle of assembly, shipment testing, transportation, unpacking, transfer, installation, teaching, operation, repacking, maintenance, and disposal, risk assessment is carried out for the life cycle of "transfer and installation" (transportation, unpacking, transfer, installation, and repacking), "use" (teaching, operation, and maintenance), and "disposal."

Table 5. Restricted Specifications Sheet

5. Person who approaches the machine		
Applicable person	Knowledge and requirements	
1) Transportation worker	Knowledge	Contents of the instruction manual
	Requirement	Person who has received training for forklifts and other transfer equipment work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
2) Unpacking worker	Knowledge	Contents of the instruction manual
	Requirement	-
3) Transfer worker	Knowledge	Contents of the instruction manual
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
4) Installation worker	Knowledge	Contents of the instruction manual, operating procedures for target robot
	Requirement	When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
5) Teaching worker Work supervisor	Knowledge	Contents of the instruction manual, operation procedures for target robot, work regulations, and robot operations
	Requirement	Person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
6) Operator	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations
	Requirement	-
7) Repacking worker	Knowledge	Contents of the instruction manual, operating procedures for target robot
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer

Table 5. Restricted Specifications Sheet (continued)

5. Person who approaches the machine		
Applicable person	Knowledge and requirements	
8) Maintenance worker Work supervisor	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations Storage procedures for target robot
	Requirement	Person who has received training for robot testing work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
9) Disposal worker	Knowledge	Contents of instruction manual, operating procedures for target robot, and waste disposal methods
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
10) People passing nearby Other workers	- Measures shall be taken to prevent entry into the fence, enclosure, or other protection surrounding the hazard zone of the robot. - Measures shall be taken to make the robot stop operation when someone approaches the hazard zone. - If another worker enters the hazard zone of the robot to perform work, robot operation shall be stopped, the drive power turned off, and a teaching worker or maintenance worker shall supervise the situation to ensure safety.	

Table 6. Restricted Specifications Sheet

6. Other

Warranty

We conduct a strict tests and inspections before delivery to ensure that the product satisfies our performance standards.
We provide warranty to cover defects of the product based on the following prerequisites.

1. Warranty period The warranty period of the product is either of the following, whichever is earlier.
   1) 18 months after delivery from our factory
   2) 12 months after installation at your site

2. Warranty scope
   1) We provide warranty for the product. Note that only the specifications and functions defined in the product specifications, catalog, and instruction manual are covered by the warranty. Any secondary or incidental damages caused by failure of the product are not covered by the warranty in any circumstances.
   2) We provide repair of a faulty product for free only if the product has been handled and used according to the instruction manual that comes with the product.
      Any repairing after the warranty period passes will be charged.

3. Disclaimer The warranty shall not apply to the following cases.
   a) Any failure or damage arising from your careless use or misuse that is not defined in the instruction manual.
   b) Any failure caused by aging or normal wear and tear (discoloration, deterioration of consumables*1).
   c) Any failure regarding sensuous phenomena such as noise not affecting the functions
   d) When any modification or disassembly without our consent is confirmed
   e) Any failure or damage arising from insufficient maintenance and inspection or inappropriate repair
   f) Any failure or damage due to act of God, fires, and other force majeure
   g) Internal data such as programs and points created or modified by users
   h) When a product purchased in Japan is taken abroad

4. Precautions
   1) SHIBAURA MACHINE shall not guarantee the standard performance of the product if you use the product beyond the specifications.
   2) If the customer does not observe the “DANGER,” “WARNING,” and “CAUTION” described in this instruction manual, SHIBAURA MACHINE will not assume responsibility for any consequential accident resulting in injury or death, damage or trouble.
   3) “DANGER,” “WARNING,” “CAUTION” and other descriptions stipulated in this instruction manual are only those which can be assumed by SHIBAURA MACHINE without predicting all events in all situations. Please be aware that the range is limited.

*1): Consumable parts refer to the replacement parts for maintenance as listed in instruction Manual “15.2 List of Replacement Parts – Robot.”

Standards, laws and regulations regarding safety of industrial robots

The international industrial standard ISO/DIS 12100 "Safety of machinery" defines measures for reducing risks regarding machinery.

Step 1: Fundamental safe design measures - Restrictions on force, speed, and energy

Step 2: Safety protection - Installation of protective fences, etc.

Additional protection measures - Installation of emergency stop devices

Step 3: Information for use - Warning labels, alerts, instruction manuals

Based on the above, the ISO/IEC defines the "Guide 51" that groups various standards in a hierarchal structure.

Safety standards for industrial robots are standardized as separate safety standards for machinery in layer C.

- ISO 10218, ISO 10218-1

After risk assessment is carried out, provision of residual risk information to the user is standardized.

- IEC 82079-1

Comply with the standards, laws and regulations in the countries where you use industrial robots.

Safety Precautions

This manual contains important information for the safe use of the robot body and the prevention of injury to the operator and others and the prevention of damage to property.
Read and understand the meanings of the following indications and symbols first, and observe these precautions.

[Meanings of Indications]

Indication	Meaning
⚠ DANGER	Indicates the imminent danger that incorrect operation could cause death or serious injury.
⚠ WARNING	Indicates the danger that incorrect operation could cause death or serious injury.
⚠ CAUTION	Indicates the possibility that incorrect operation could cause injury[1] to operator.
NOTICE	Indicates the possibility that incorrect operation could cause property damage[2] and advice about performing work.

1) Injury indicates injuries such as wounds, burns, and electric shock that do not require hospitalization or long-term outpatient treatment.
2) Damage to property indicates extensive damage related to the destruction of assets or materials.

NOTICE
• In order to install and dispose of the robot safety, be sure to read the “Safety Manual” Instruction Manual provided separately thoroughly before actually starting work.

Service Network

Please refer to the materials distributed from the SHIBAURA MACHINE agent.

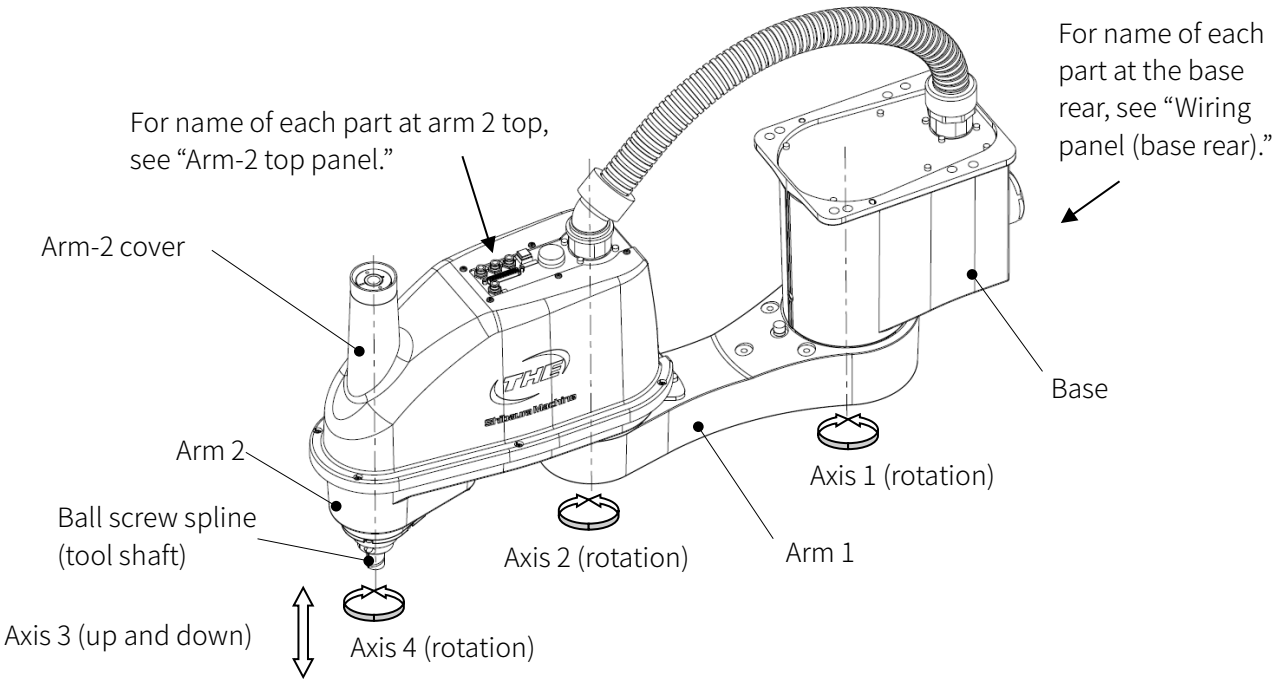
Specifications

This part describes the basic specifications, name of each part, coordinate system, and external dimensions of the robot.

1. Specifications

1.1 Name of Each Part

Fig. 1.1 shows the name of each part of the robot.



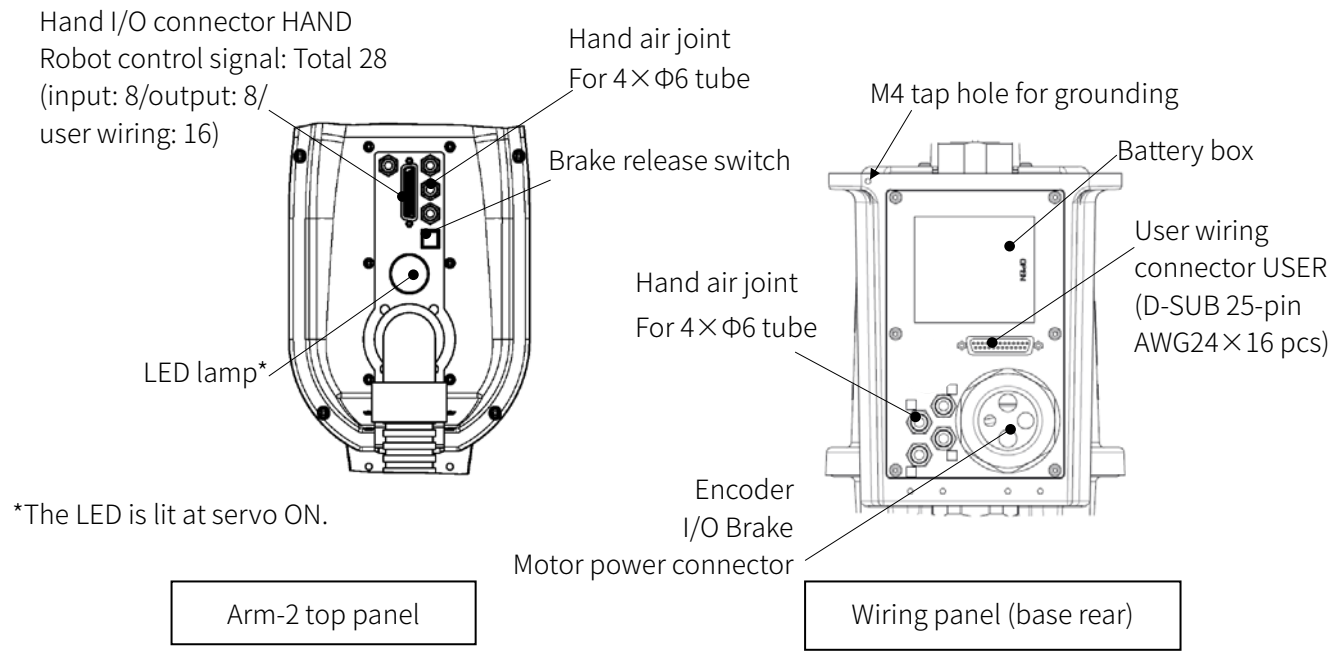


Fig. 1.1 Name of each part

1.2 Coordinate System

This is identical to the THE600-E. For details, see “1.2 Coordinate System” in the “STEA1211: THE600-E Robot Manual”.

1.3 External Dimensions

Fig. 1.2 refers to the external dimensions of the THE600-T-E.

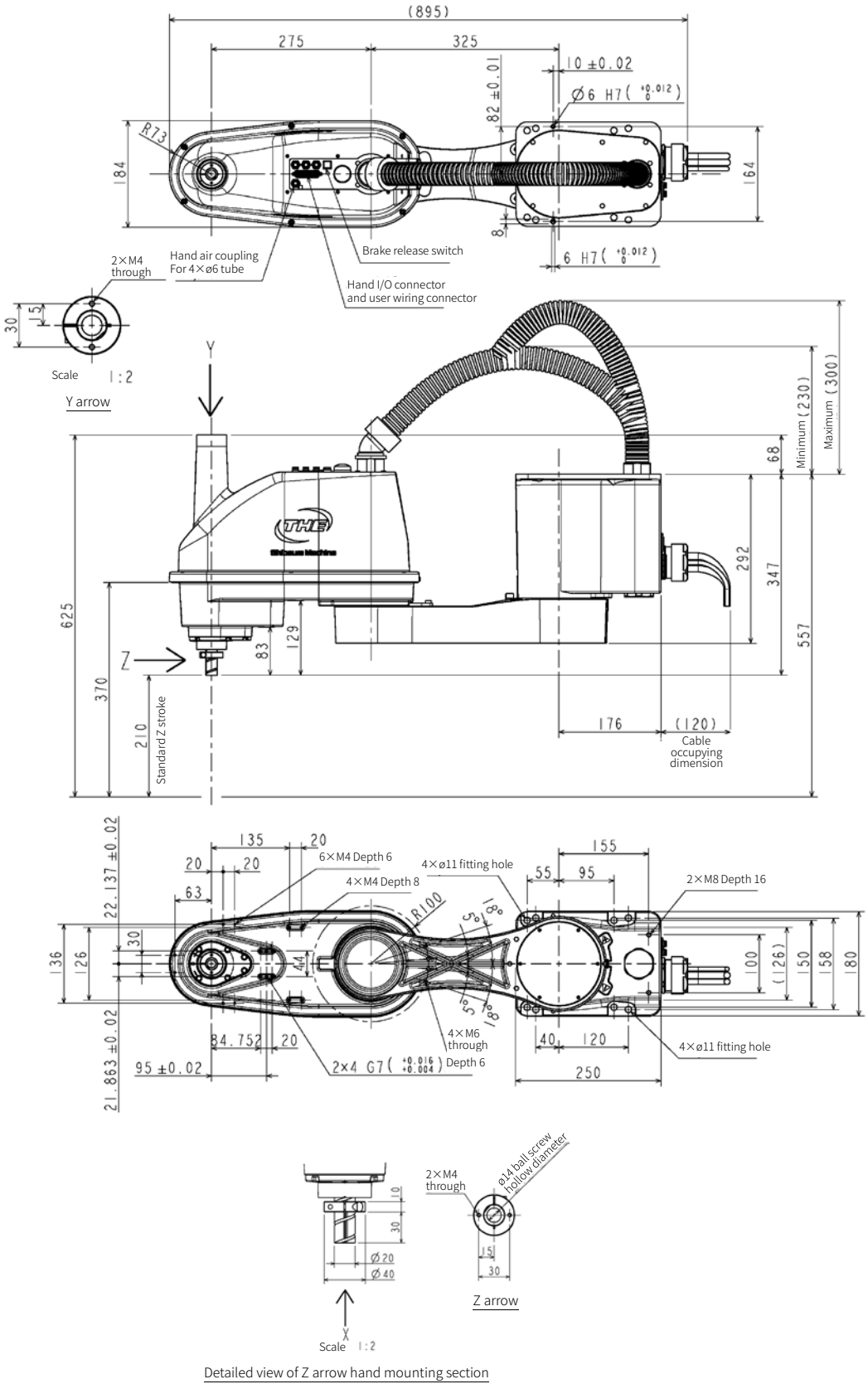


Fig. 1.2 External dimensions of the robot (THE600-T-E)

1.4 Specifications Table

Item		Specifications
Structure		Horizontally articulated type SCARA robot
Model		THE600-T-E
Applicable controller		TS5000-EMS
Mass of robot body		31 kg
Number of controlled axes		Four (4)
Arm length		600 mm (275 mm + 325 mm)
Motor capacity/current limit	Axis 1	1000 W / 28.2Ao-p
	Axis 2	400 W / 14.1Ao-p
	Axis 3	400 W / 18.4.Ao-p
	Axis 4	400 W / 14.1Ao-p
Operating range	Axis 1	±140 deg
	Axis 2	±150 deg
	Axis 3	0 to 210 mm
	Axis 4	±360 deg
Maximum speed *1	Axis 1	457 deg/s
	Axis 2	672 deg/s
	Axis 3	2000 mm/s
	Axis 4	2359 deg/s
	Composite speed of axes 1 and 2	8.0 m/s
Rated payload mass*1		2 kg
Maximum payload mass*1		12 kg
Permissible load inertia*1		0.25 kg·m^2
Maximum offset of load gravity center		100 mm
Repeatability*2	Composite speed of axes 1 and 2	±0.01 mm
	Axis 3	±0.01 mm
	Axis 4	±0.005 deg
Cycle time (when payload mass is 2 kg)*3		0.31 s
Drive system		AC servo motor
Position detection method		Absolute
Wiring for hand		Input: 8, output: 8
Paint color*4 Material color*4		Arm 2: PANTONE 293C or equivalent Base and arm 1: White AN-90 or equivalent Arm cover: White AN-90 or equivalent
Noise level*5		73 dB (A)
Power supply capacity*6		4.3 kVA

*1: The speed and acceleration are limited in accordance with the operation pattern, the load mass, and the offset value.

*2: Repeatability is the repeat accuracy in one direction at a constant ambient temperature and machine temperature. It is not absolute positioning accuracy. It may exceed the specification values depending on the operation pattern, load weight, and offset value. As for X-Y and C, they represent values at the upper limit of Z. Trajectory accuracy is not guaranteed.

*3: Continuous operation with standard cycle operation pattern and exceeding the effective load ratio cannot be performed.
During shuttle for rough positioning in horizontal direction of 300 mm and vertical direction of 25 mm.

*4: The color may differ according to each production lot. Be aware that there is no problem in the very nature of the product quality.

*5: The measurement conditions are as follows.
Operating conditions: Payload 2 kg, simultaneous movement of all joint axes, maximum speed, and maximum acceleration
Measurement position: Distance of 1 m from the 1st axis of the robot in five directions (front, rear, right, left, and up)
Maximum sound pressure position: 1 m directly above the 1st axis of the robot

*6: The measurement conditions are as follows.

Operating condition:	Rated load and 2 kg, concurrent operation of all axes, maximum speed, and maximum acceleration
Measuring instrument:	Power analyzer (POWER ANALYZER 3390)

NOTICE

- Never use the robot in the high-temperature environment.
  Otherwise, the performance and lifespan of the alkaline batteries used for the position detector batteries will be reduced.
  - If the robot will be used in a high-temperature environment, please consult with the SHIBAURA MACHINE sales office.
- Never allow the lowered ball screw spline shaft to collide with an obstacle.
  Damage will occur before an alarm is triggered.
- When micro-vibrations occur, be sure to decelerate the robot.
  Otherwise, the performance and lifespan of the robot may be reduced. The robot will malfunction.
  - Use the ACCEL/DECEL command to adjust to an acceleration where the robot will not vibrate.
  - Micro-vibrations may occur on the robot tip depending on the orientation of the robot.
- To operate the 1st, 2nd, and 4th axes, be sure to raise the ball screw spline shaft as high as possible.
  If it is lowered, the shaft could be damaged prematurely.
  - To operate the 1st, 2nd, and 4th axes with the ball screw spline shaft lowered, use the SPEED command and ACCEL/DECEL command to adjust the operating speed and acceleration to prevent the ball screw spline shaft from vibrating.

2. Functions

Functions is the same as that for the THE400-E. For details, see the "STA1213 THE400-E Robot Manual."

Installation

This part describes information you should know before unpacking, transporting, installing, and operating the robot.

3. Transportation

The worker should receive safety training defined by each country's laws and regulations. Be aware that the failure and accident resulting from the work done by the customer will not be under our warranty.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	

3.1 Transporting by Forklift

⚠ WARNING

- Never carry the packing boxes by hand when unloading them from a truck or other vehicle.
  A packing box could fall on a worker.
  - Use a forklift or similar equipment.
  - Do not allow people to approach the transportation route.

NOTICE

- Never carry the packing boxes by hand when unloading them from a truck or other vehicle.
  The worker may be unable to support the packing box and could fall over, resulting in damage to the robot.

The robot is packed and shipped in cardboard boxes. Unload the packing boxes stacked in the truck or other transportation equipment, and carry them to the unpacking location or other site. Transport carefully to ensure that the robot is not damaged.

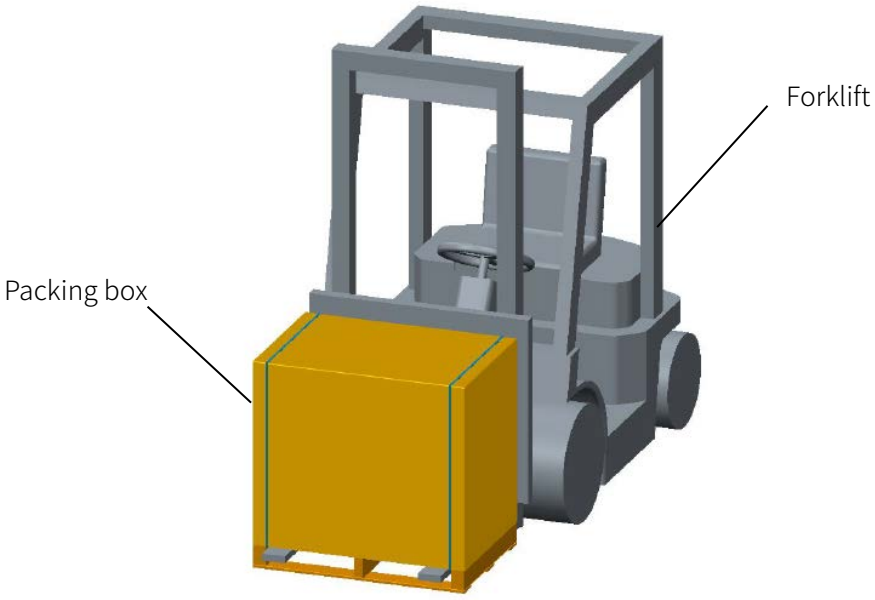


Fig. 3.1 Example of Transport Method for Packing Boxes

Table 3.1 Size and Mass

Packing size	630mm（W）×900mm（H）×920mm（D）
Packing weight	About 66 kg (not including options)

4. Unpacking

The worker should receive safety training defined by each country's laws and regulations. Be aware that the failure and accident resulting from the work done by the customer will not be under our warranty.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	

⚠ CAUTION

- Never touch fluid that has leaked from the position detector backup batteries.
  If any fluid gets on your skin or in your eyes, inflammation will occur.
  - Do not bring the robot into high-temperature environments because fluid leakage is more likely to occur.
  - Wear protective gloves if there is a possibility of coming into contact with leaked fluid.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Replace any parts that the fluid has splashed on.

4.1 Unpacking

Fig. 4.1 shows the packing state. Open the packages in a location easily accessible, where the equipment is to be installed. Take careful precautions not to damage the robot. After opening the packages, make sure that all the accessories are present and that nothing has been damaged during transport. For robot and controller accessories, see the accessory list packed with the controller.

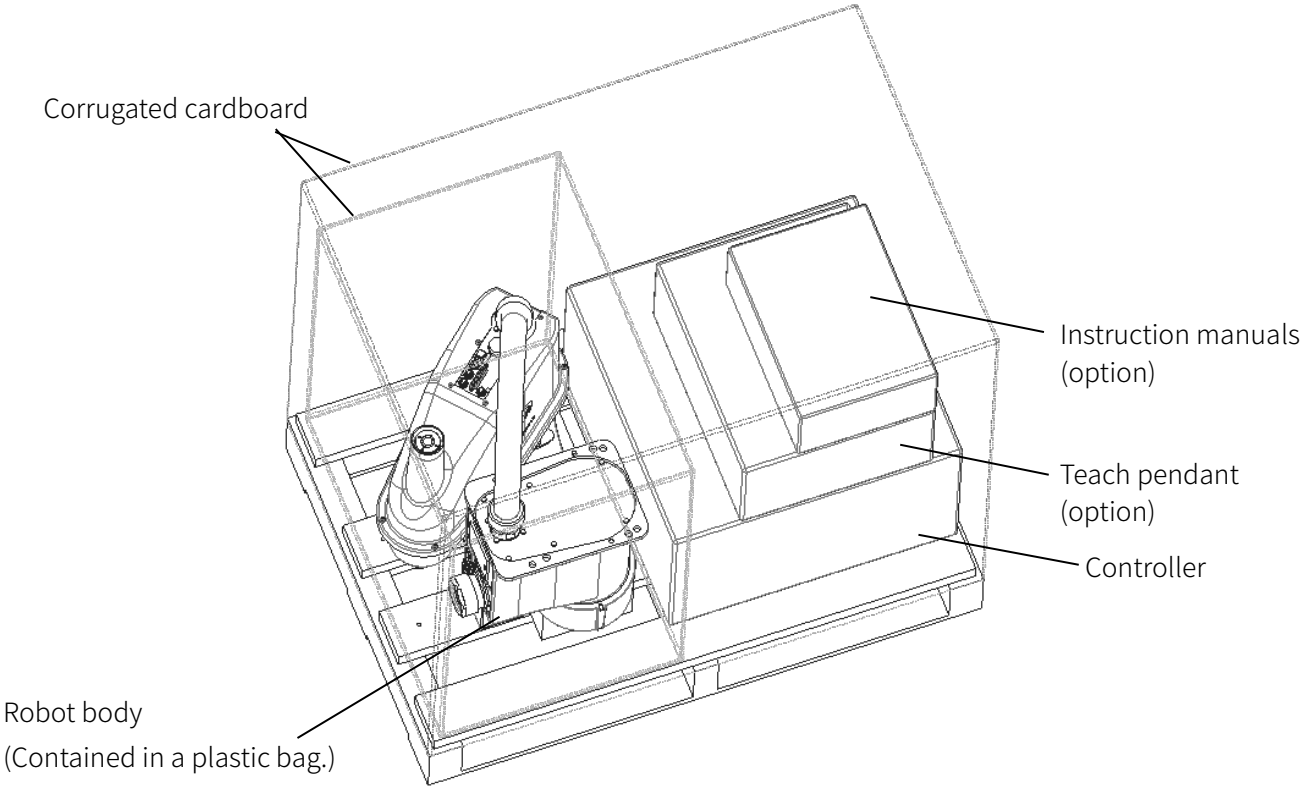


Fig. 4.1 Packaging state (THE600-T-E)

Table 4.1 Size and Mass

Packing size	630 mm (W) × 900 mm (H) × 920 mm (D)
Packing mass	About 66 kg (option not included)

⚠ CAUTION

- When cutting the packing strap with the cutter, never point the blade edge towards a person. Otherwise, the person could get cut by the cutter.
  - Use a nippers or scissors.
  - Wear protective gloves.

⚠ DANGER

- After the work is completed, be sure to always check that the robot is not damaged.
  If the power is turned on for a malfunctioning robot, the robot could hit the worker due to an unexpected movement.
  - Check the appearance based on the external view shown in the manual.
  - Check parts based on the accessory list attached to the controller.
  - If damage is found or the parts are insufficient, please contact SHIBAURA MACHINE Sales Division.

NOTICE

- After the work is completed, be sure to always check that the robot is not damaged.
  A damaged robot could malfunction.
  - Check the appearance based on the external view shown in the manual.
  - Check parts based on the accessory list attached to the controller.
  - If damage is found or the parts are insufficient, please contact SHIBAURA MACHINE Sales Division.
- Be sure to dispose of any unneeded packing materials in accordance with the customer's company regulations.
  When storing the packing materials, be sure to manage them to prevent them from being used for other purposes than packing the robot.

MEMO

When storing the robot after unpacking or repacking for transportation, the customer should obtain a new packing strap (PP band), and secure the product in place as shown in Fig. 4.1.

⚠ WARNING

- Be sure to secure the robot to be stored in a packing box as it was delivered.
  Otherwise, the robot could fall over.
  - The customer must obtain new packing straps (PP bands).
  - If no packing box is available, use the mounting holes in the base to secure in place.

MEMO

- *When taking out the robot, refer to 5 Transfer.*
- *Refer to the separate controller manual for information on transporting, unpacking, and carrying the controller packing box included in the packing box.*

4.2 Storage

Transfer to the storage site carefully so as to ensure that the robot is not damaged. Avoid storing the robot and controller without using them for long periods of time after unpacking them.
Also, select a storage location that meets the environmental requirements in 6.1Installation Environment.

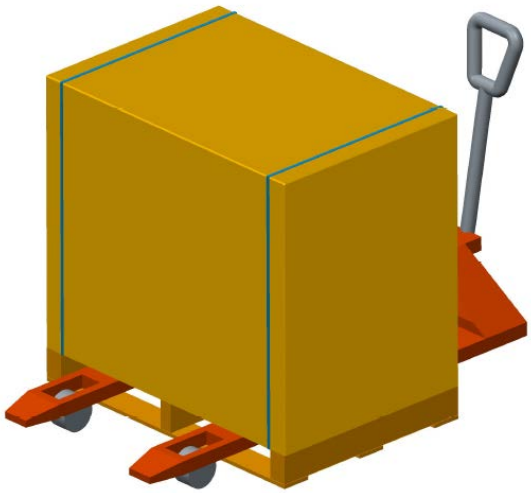


Fig. 4.2 Example of transfer method when storing

⚠ WARNING

- Never carry the packing boxes by hand.
  A packing box could fall on a worker.
  - Use a hand lifter or similar equipment.
  - Do not allow people to approach the transportation route.

NOTICE

- Never carry the packing boxes by hand.
  The worker may be unable to support the packing box and could fall over, resulting in damage to the robot.
  - Use a hand lifter or similar equipment.
  - Do not allow people to approach the transportation route.

Avoid storing the robot and controller for long periods of time after unpacking them.
If this is unavoidable, be sure to strictly observe the following storage precautions.

NOTICE

- Never store the robot at a location exposed to direct sunlight, high temperature, or high humidity. Otherwise, the cable ducts, timing belt, and resin parts could deteriorate.
- DO NOT store the robot at a place where it is exposed to direct rain or water splashes, or at a place containing any toxic gas or liquid. Otherwise, the parts could deteriorate prematurely.
- When storing the robot, seal it in a plastic bag, and put a desiccant inside the bag.
- This prevents the entry of small animals and prevents rust from forming and dust from adhering to the ball screw spline unit.
  - Before storing the robot, apply grease to the ball screw spline shaft.
  - For the application method, see “STEA1211 THE600-E robot manual.”
  - Also apply grease to the ball screw spline shaft before use.
- Before operating a robot that was stored, be sure to perform operation at low speed (about 20% of override) for about four hours.
  During storage, the grease viscosity increases. If the robot is suddenly operated at a high speed, the poor lubrication could result in failure.
- Before operating a robot that was stored, be sure to replace the position detector batteries. (See “STEA1211 THE600-E robot manual”)
  An alarm occurs before periodic inspection.
  - If power is not supplied to the robot for an extended period of time due to storage or other reasons, the data retention period by the position detector backup batteries will be shorter.
- When storing the robot, use the supplied fixtures to secure the arm and base, and then firmly secure the mounting sections. If placed directly on the floor, the robot will be unstable and will fall over.

5. Transportation

The worker should receive safety training defined by each country's laws and regulations. Be aware that the failure and accident resulting from the work done by the customer will not be under our warranty.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	

⚠ WARNING

- Never use an awkward posture when taking out the product from the packing box.
  Overexertion could injure the worker's back.
  - Provide space where work can be performed easily.
- Be sure to always store floor-routed cables in ducts and similar protection.
  Otherwise, someone could trip and fall on them.
  - Workers should wear a helmet to protect their head.

⚠ CAUTION

- Never touch fluid that has leaked from the position detector backup batteries.
  If the fluid gets on the skin or eyes, it can cause inflammation.
  - Do not bring the robot into high-temperature environments because fluid leakage is more likely to occur.
  - Wear protective gloves if there is a possibility of coming into contact with leaked fluid.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Replace any parts that the fluid has splashed on.

5.1 Transportation

Transfer the robot while being careful that it is not subjected to any impacts or vibrations. If the robot will be subjected to vibrations over a long period of time, be sure to fully tighten all the fixing jigs and base set bolts.

In principle, the robot should be conveyed in the state shown in Fig. 5.1.
The arm is folded and is secured by the supplied fixing jig. (The robot is packed in this state, and even after it is unpacked, it should be transferred in this state.) When securing in place, be careful not to apply a large force to the ball screw spline shaft.

5.2 Mass and Dimensions

Fig. 5.1 shows the mass and external dimensions of the robot.

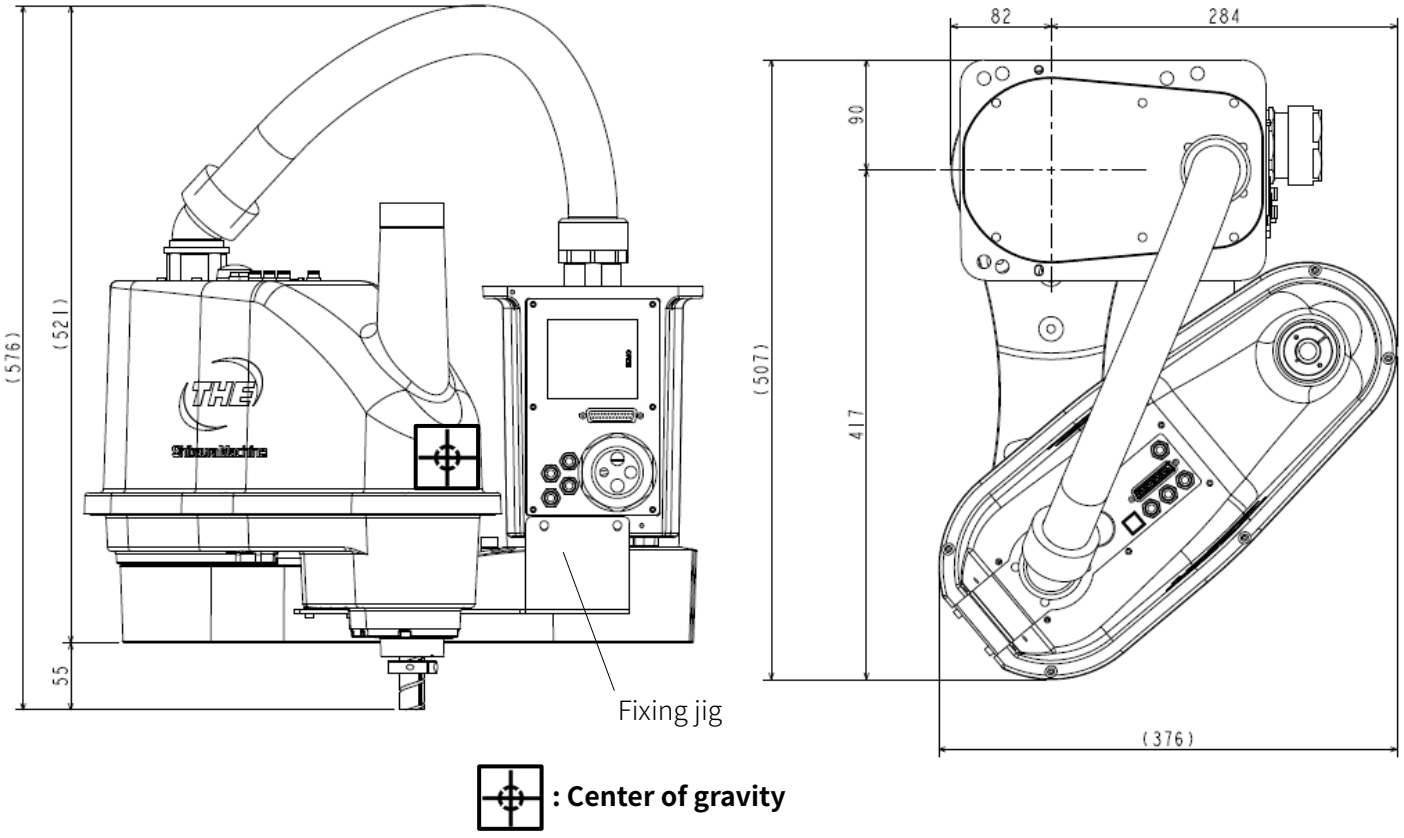


Fig. 5.1 Outer dimensions at transport (THE600-T-E)

⚠ CAUTION

- When transferring the robot, be sure to always secure in place using the supplied fixing jigs. Otherwise, a worker could get caught in an arm that moved.
  - Check that no bolts of the jig are loose.
  - The weight of the arm section is approximately 17 kg.
  - Wear protective gloves to prevent slippage and to protect the hands.

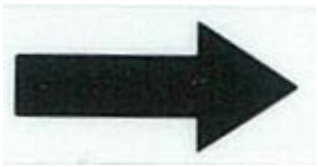
5.3 ransporting the Robot by Crane

⚠ WARNING

- When hoisting the robot, be sure to always follow the following safety guidelines.
  Otherwise, the robot could drop.
  - Use the **supplied** eyebolts or eyebolts with a working load of 0.79 kN or more when suspended at 45°.
  - Use a wire with a load limit that can sufficiently bear the weight of the robot.
  - Be sure to always hoist and lower the robot slowly.
  - Clear a transfer route, and make sure that no one goes under the suspended load.

The robot can be hoisted and transferred.

The following arrow marks are affixed to the base and 1st arm of the robot. Mount the two supplied M8 eyebolts into the tapped holes indicated by the arrow marks as shown below in Fig. 5.2, secure the wire to the eyebolts, and slowly hoist the unit by two-point suspension.

Label	Label description
	- Mount the supplied eye bolts into the tapped holes indicated by the arrow mark.

- When hoisting and transferring the robot by crane, be sure to mount the supplied eyebolts into the tapped holes indicated by the arrow marks. Otherwise, the robot will be unstable and could crash into a worker.

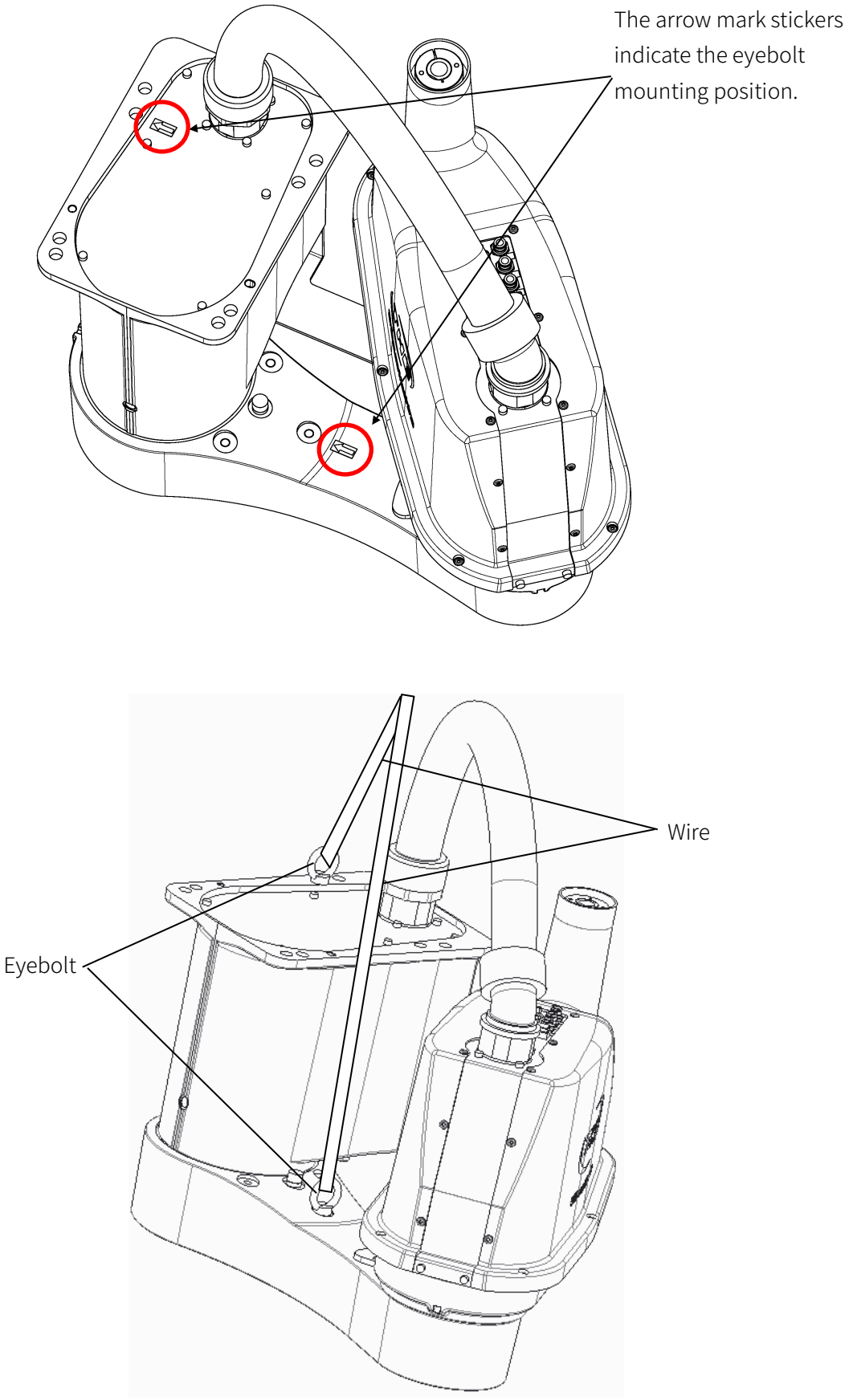


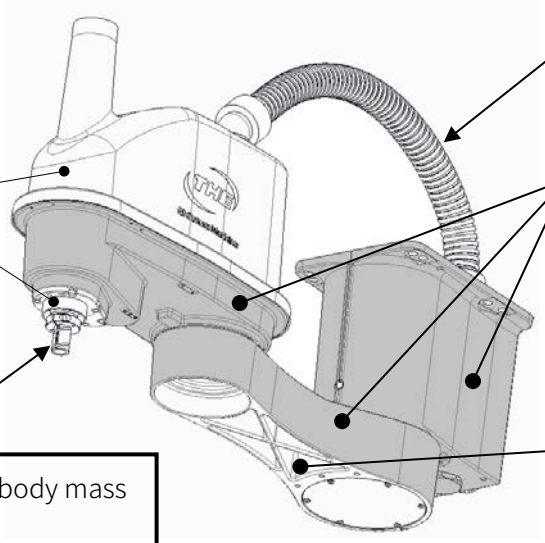
Fig. 5.2 Lifting up the robot

NOTICE

- When hoisting and transferring the robot, mount the supplied eyebolts into the tapped holes indicated by the arrow marks, and use the eyebolts to hoist the robot. Otherwise, the robot could lose its balance and fall over.
  - When hoisting with eyebolts provided by the customer, select a product with a working load of 0.79 kN or more when suspended at 45°.
- Never touch the ball screw spline shaft with your bare hands.
  Otherwise, rust could form prematurely.
  - Wear gloves whenever touching the ball screw spline shaft.
  - If the ball screw spline shaft is inadvertently touched with your bare hands, apply grease to the ball screw spline shaft.
- Be sure to hoist and lower the robot slowly.
  Otherwise, the robot could collide with the peripheral equipment, frame, or other objects, resulting in failure or damage.
- After installation is completed, be sure to remove the fixing jig and eyebolts.
  If the robot is moved while they are mounted, the robot could be damaged or fail.
  - The removed eyebolts will be used when the robot is transferred again, and so keep them in a safe place for future use.

5.4 Transferring the robot by Workers

If lifting up the robot by workers, see Fig. 5.3 to hold the robot by hands.



MEMO

This figure shows the handing areas, and does not show the conveyance posture. For the conveyance posture, see Fig. 5.1

Fig. 5.3 Handling areas of the robot (THE600-T-E)

⚠ WARNING

- The robot should never be transferred by only one worker.
  Otherwise, the robot cannot be fully supported and could drop.
  - The robot weight is 31 kg.
  - Lift the robot using more than one person.
  - Use a cart or similar device to the transfer location.
  - Clear a transfer route, and make sure that a dropped robot would not hit anyone.
  - Wear protective gloves and safety shoes.

NOTICE

- The robot should never be transferred by only one worker.
  Otherwise, the worker may be unable to support the robot and it could fall over.
  - Do not use a product that has been dropped.
- Never touch the ball screw spline shaft with your bare hands.
  Otherwise, rust could form prematurely.
  - Wear gloves whenever touching the ball screw spline shaft.
  - If the ball screw spline shaft is inadvertently touched with your bare hands, apply grease to the ball screw spline shaft.
- When lifting up the robot by workers, be sure to hold the gray portions shown in Fig. 5.3 by hands.
  If other sections are held, the held sections may be damaged. This could apply excessive force to the mechanical section, resulting in a breakdown.
- After installation is completed, be sure to remove the fixing jig.
  If the robot is operated while the jig is mounted, the robot may be damaged or fail.

6. Installation of the Robot

The worker should receive safety training defined by each country's laws and regulations.
Be aware that the failure and accident resulting from the work done by the customer will not be under our warranty.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	

MEMO

- *Before starting work, check the information in STEA1211 THE600-E robot manual 10.1 Rescuing a Person Trapped by a Robot, and conduct a rescue drill.*
- *The 1st, 2nd, and 4th axes of THE600 can be moved by turning off the drive power (servo off) and pushing them by hand. By releasing the brake while the 3rd axis is powered on, it can be pushed and moved by hand without turning on the drive power supply (turning on the servo). For details, see STEA1211 THE600-E robot manual 9 Teaching.*
- *We have prepared various methods for guiding the robot, which is done by turning on a drive power supply (turning on the servo). For details, refer to the following the separate manuals.*
  - TP5000 Operation Manual
  - TCmini (Internal PLC) Manual
  - Communication Manual
  - TSAssist User's Manual

6.1 Installation Environment

Table 6.1 shows the environmental conditions for the location in which the robot are to be installed.

Table 6.1 Environmental conditions for robot

Item	Specifications
Temperature	During operation: 0 to 40°C During transport and storage: -10°C to 50°C
Humidity	During operation, transport, and storage: 20% to 80% (no condensation) DO NOT store or install the equipment in locations where it will be exposed to water or other fluids.
Altitude	1000 m or less
Vibrations	During operation and storage: 0.98 m/s^2 or less
Dust	The environment must not be exposed to inductive dust. Consult with SHIBAURA MACHINE beforehand if the equipment will be used in a dusty environment.
Gas	The environment must not be exposed to corrosive or combustible gas.
IP (Ingress Protection) rating	IEC60529-1:1989, IP2X
Overvoltage category	IEC60664-1:2020, Class III
Protection against electric shock	IEC61140:2016, Class I
Pollution degree	IEC60664-1, Pollution degree 3
Sunlight	The environment must not be exposed to direct sunlight.
Power noise	No sources of excessive noise must be in the surrounding area.
Magnetic field	No devices that generate strong magnetic fields must be in the surrounding area
Other ambient environmental requirements	The environment must not be exposed to iron powder, oil, salt, or organic solvents.

⚠ DANGER

- Water and other liquids must not be brought into the area around the robot.
  If a liquid is spilled, it could result in an electric shock.
- Do not put devices that generate excessive amounts of noise near the robot.
  Otherwise, a robot could malfunction due to noise and collide with a worker.

⚠ CAUTION

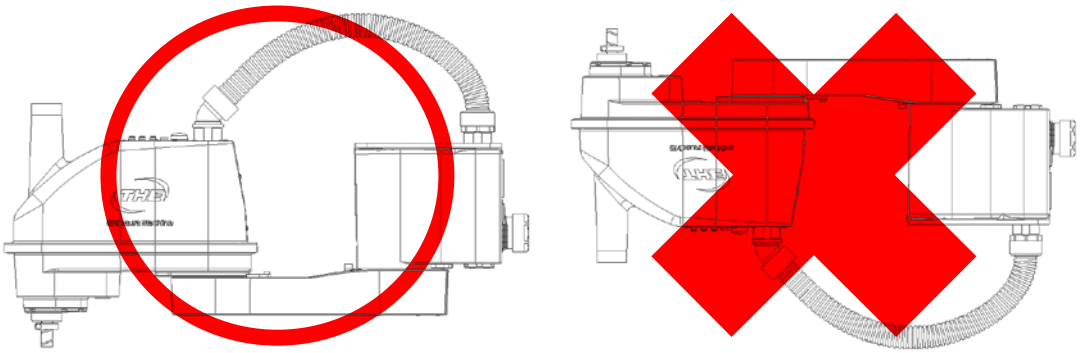
- Never touch fluid that has leaked from the position detector backup batteries.
  If any fluid gets on your skin or in your eyes, inflammation will occur.
  - Do not bring the robot into high-temperature environments because fluid leakage is more likely to occur.
  - Wear protective gloves if there is a possibility of coming into contact with leaked fluid.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Replace any parts that the fluid has splashed on.

NOTICE

- Never turn on the controller in a location where it will be exposed to water to a degree that exceeds the protection class, where it will be exposed to liquids other than water, or where the site is humid enough to cause condensation.
  Otherwise, the robot could fail.
- When operating the robot after an extended stop at a low temperature (10°C or less), be sure to perform operation at a low speed (about 20% of override) for four (4) hours.
  The grease viscosity increases in low-temperature environments. If the robot is suddenly operated at a high speed, the poor lubrication could result in failure.
  - Apply grease to the ball screw spline shaft before use.

6.2 Installation

Before the robot can be installed, the installation layout must be studied in consideration of the working envelope, the coordinate system, and the maintenance clearance.



MEMO

Usage of this robot in an orientation other than the ceiling mount orientation is prohibited.

⚠ WARNING

- Be sure to always store floor-routed cables in ducts and similar protection.
  Otherwise, someone could trip and fall on them.
  - Workers should wear a helmet to protect their head.
- Never modify parts except those that are expressly permitted.
  Otherwise, the robot could fail, and cause an expected accident.

⚠ CAUTION

- Be sure to connect the cables to the controller after the robot is installed.
  Otherwise, a worker could receive an electric shock due to touching a cable that was broken due to a carried object that was dropped.
  - If a cable is broken, do not turn on the power.
- When removing the fixing jig, never grasp the edges strongly.
  If excessive force is applied, a hand or finger could be cut on the fixing jig.
  - Wear protective gloves.
  - The bolts that are fastened to the robot may become hardened due to vibration during transportation or transfer.

NOTICE

- Be sure to route the cables after installation is completed.
  Otherwise, a carried object could drop on a cable that was already routed, resulting in breakage.
  - Store the wiring in ducts or other protection.
- The cable between the robot and controller is not a movable cable. If the robot is mounted on a traveling axis or similar structure, the cable will be damaged prematurely.
- Never modify parts except those that are expressly permitted.
  Otherwise, the performance and lifespan could be reduced and the robot could malfunction.

6.2.1 Installing the Robot

⚠ WARNING

- Never install the robot on a frame that is unable to support the load during operation.
  Otherwise, the robot and frame could fall over together during operation.
  - Prepare the frame by referring to the load applied during horizontal operation (Table 6.2).

NOTICE

- Never install the robot on a frame that is unable to support the load during operation.
  Otherwise, the robot could vibrate during operation. The robot will malfunction.
  - Prepare the frame by referring to the load applied during horizontal movement (Table 6.2).
- Be sure to install the robot on a level surface.
  Otherwise, the performance and lifespan of the robot could be adversely affected. The robot will malfunction.
- Be sure to secure the frame where the robot is installed to a floor or wall.
  Otherwise, the frame will move due to the force applied during robot operation.
- The robot should never be installed by only one worker.
  Otherwise, the worker may be unable to support the robot and it could fall over, resulting in breakdown.

The robot is secured by using the mounting holes on the base (four (4) places).
Use attached hexagon socket head cap screws M10×25.
Table 6.2 lists the load on the frame during horizontal movement, and Fig 6.1 shows the robot installation dimensions. The pinholes are provided on the base unit. If the robot position is to be aligned in the base coordinate system or the robot must be replaced, the robot can be positioned by the pinholes.

Table 6.2 Load on the frame during horizontal movement

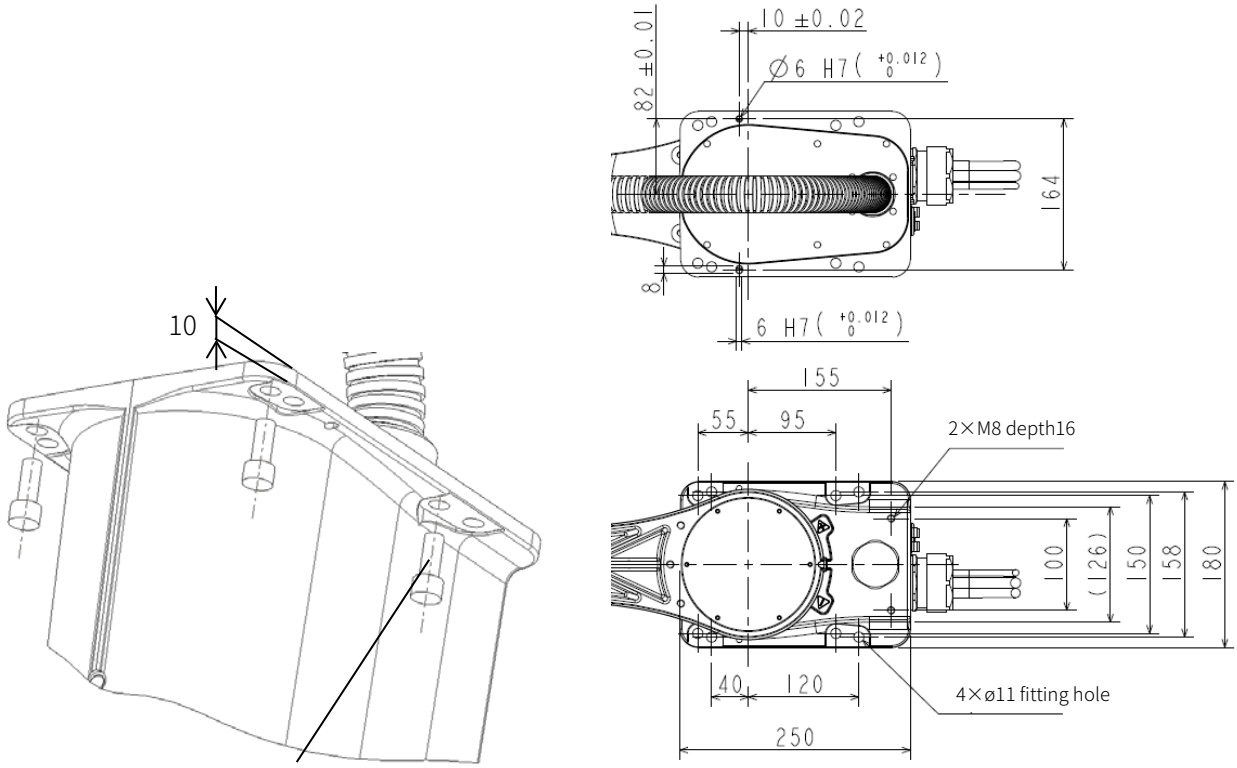
Model	Load on the frame during horizontal movement (Nm)	Robot body mass (kg)
THE600-T-E	650	31

*This value is a reference value. When designing the frame, consider safety factor.

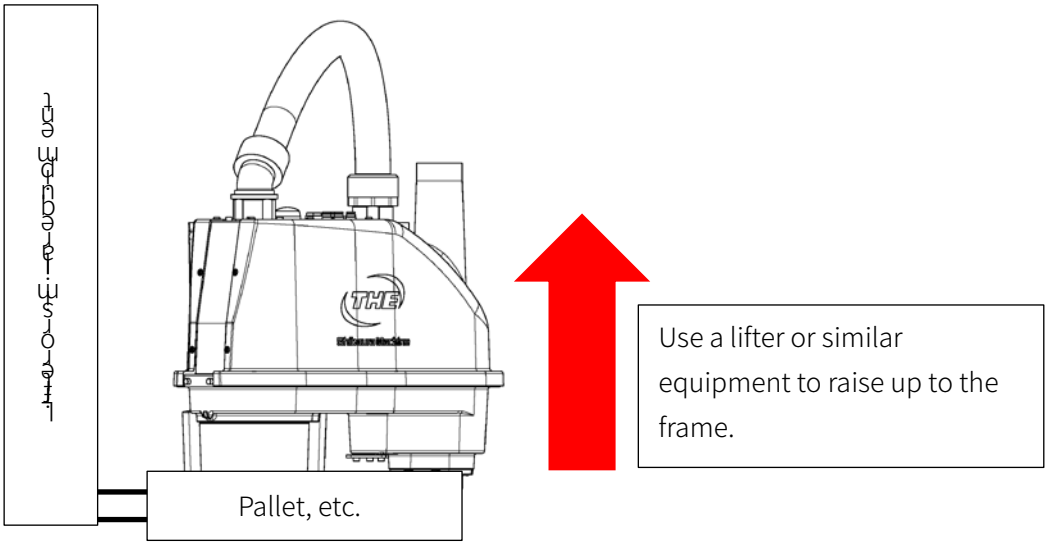
Use the frame that has the flatness and surface roughness listed in Table 6.3 for the frame on which the robot is to be installed.

Table 6.3 Flatness of robot installation part of the frame

Flatness of robot installation surface	Surface roughness of robot installation surface
0.02	Ra 6.3



Hexagon socket head cap screw M10×25

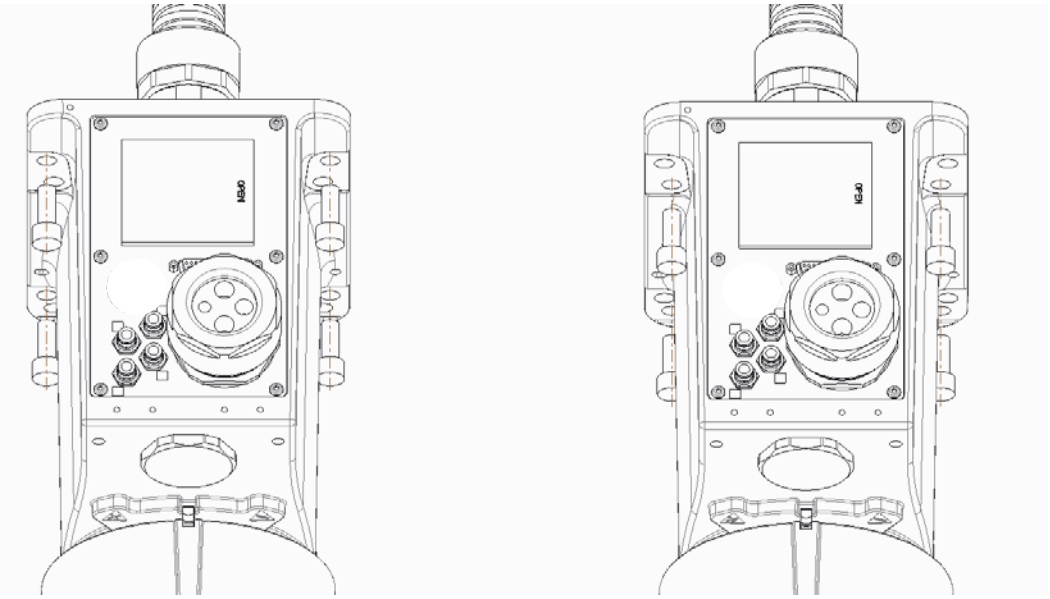


MEMO

Place the robot on a pallet or similar surface, and then use a forklift or similar equipment to transport to the installation frame.

Fig 6.1 Setting method (THE600-T-E)

The robot base has a total of eight mounting holes for installation. Use four of eight mounting holes to install the robot on the frame. The combination examples of the mounting holes are shown in Fig. 6.2.



<Example 1> <Example 2>

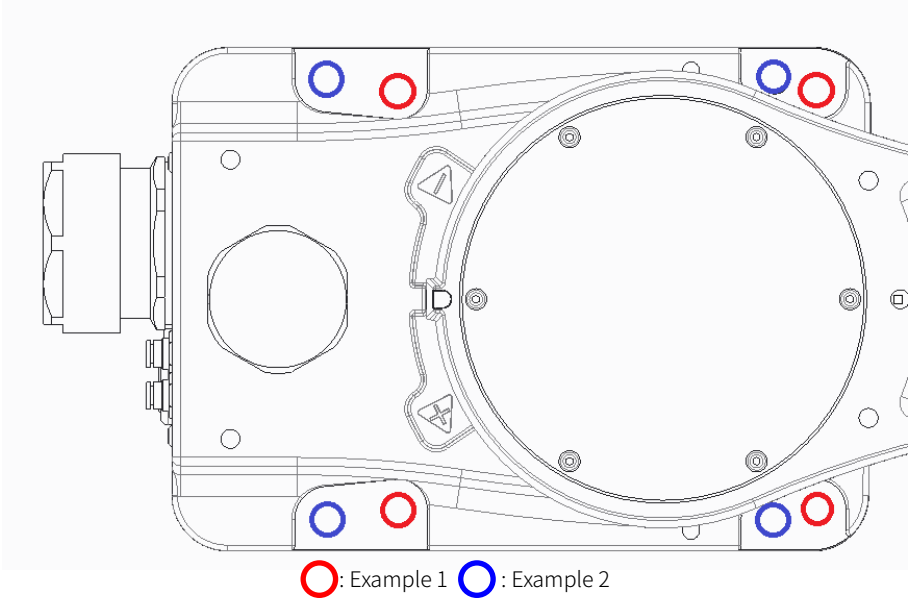


Fig. 6.2 Combination examples of mounting holes

6.2.2 Grounding the Robot

As shown in Fig. 6.3, the robot has the tap hole for grounding. When grounding the robot, use this tap hole to secure the grounding wire terminal with the bolt.

⚠ DANGER

- Before turning on the power, be sure to always check that the ground wire is connected.
  Otherwise, an electric shock due to a leakage current could occur when touching the robot.

NOTICE

- Be sure to always connect a ground wire.
  Otherwise, the robot could fail due to noise.

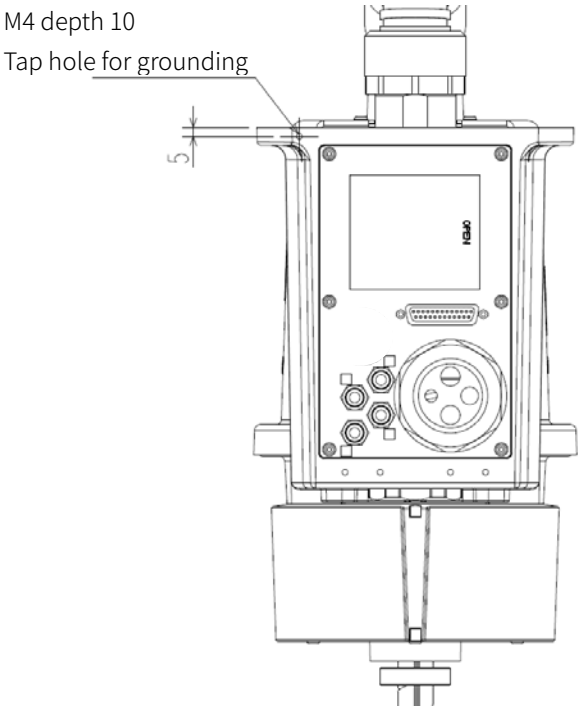


Fig. 6.3 Tap hole for grounding

MEMO

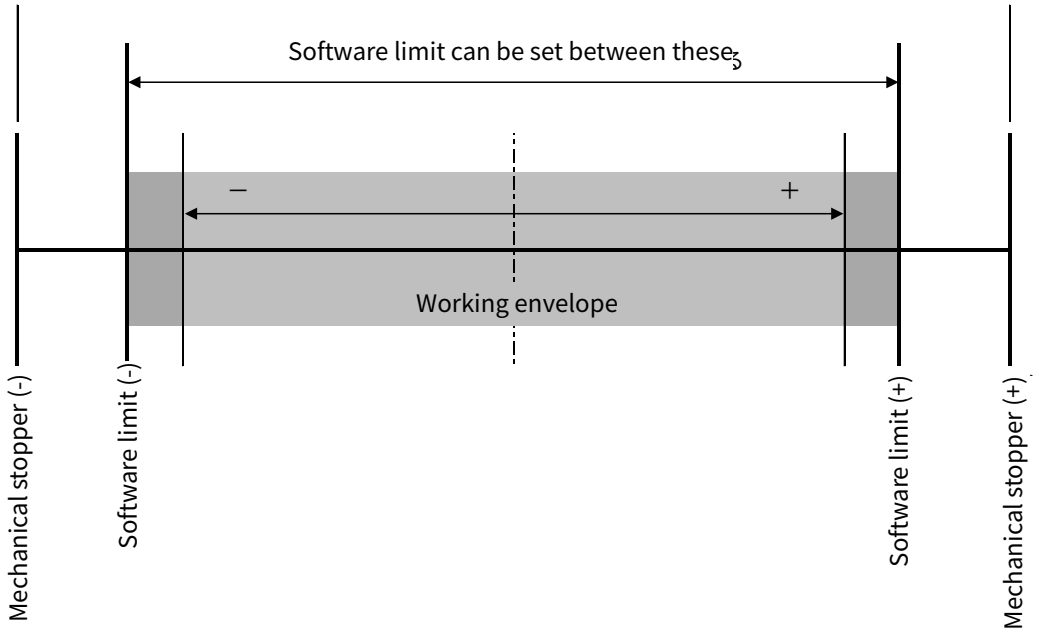
For the connecting ground wire, use a 2.0 mm2 thick cable with a combination of green and yellow wires, and connect with a cable length not exceeding 1 meter.

6.3 Operating Range

Fig. 6.4 shows the operating range.

Each joint axis can operate within the working envelope. To prevent the robot from operating outside the working envelope due to mistaken operation, it has two types of movement limit functions. Additionally, there are software limits that the user can set up.

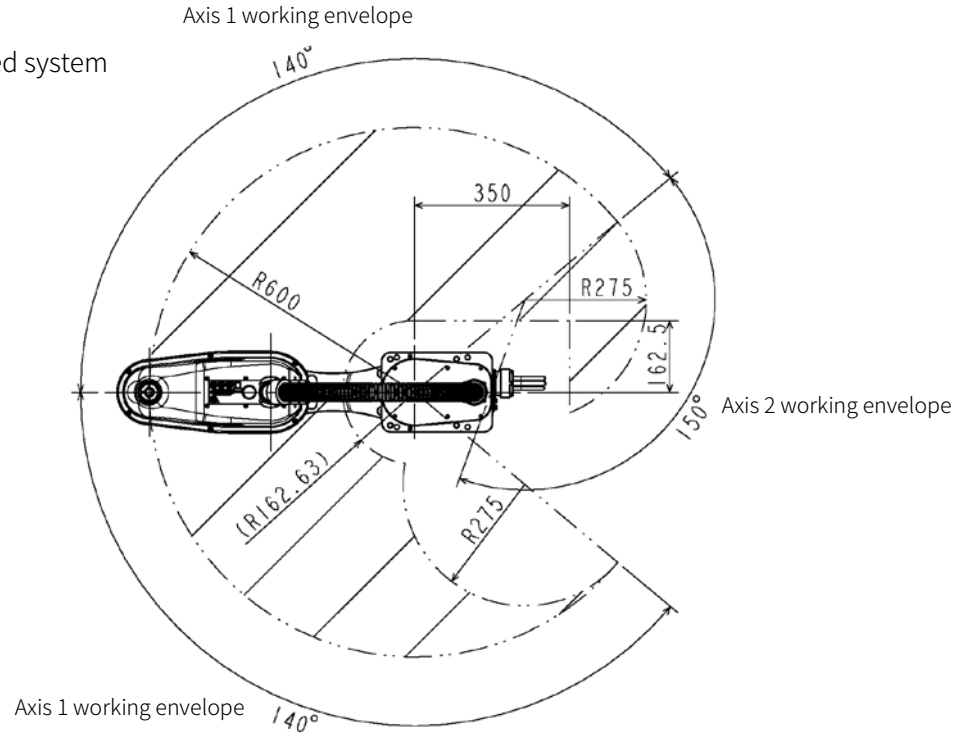
For details, refer to the User Parameters Instruction Manual provided separately.



MEMO

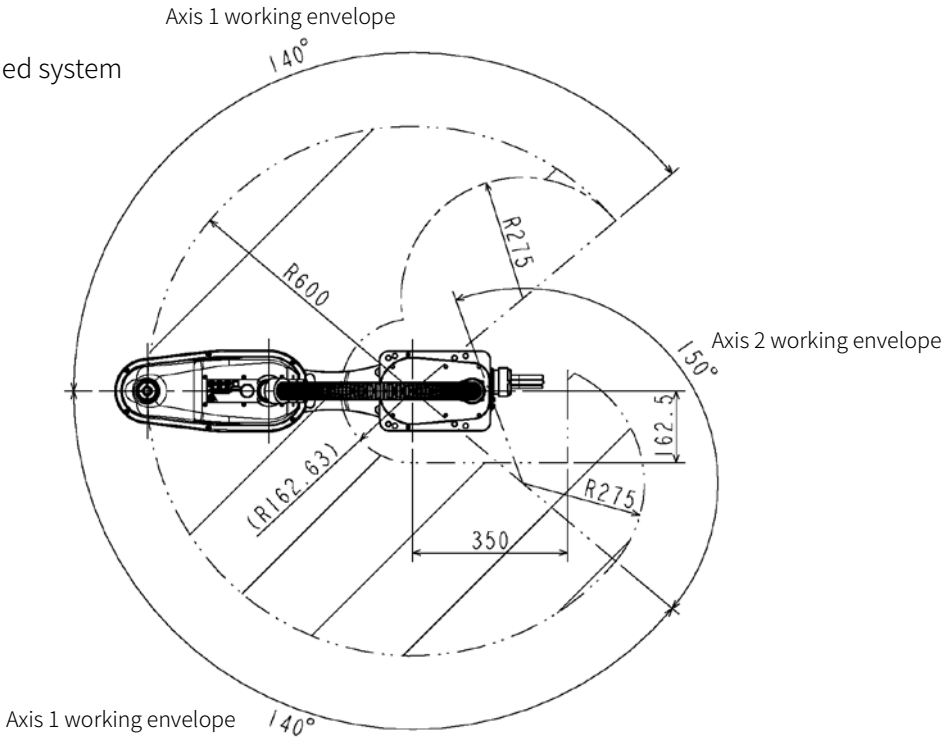
The software limit cannot be set in the range that is wider than the value set at shipment. The software limit can be set in the range that is narrower than the value set at shipment. So, to change the mechanical stopper position and narrow the working envelope, be sure to change the software limit as well.

For left-handed system



Left-handed system working envelope

For right-handed system



Right-handed system working envelope

Fig. 6.4 Operating range

6.3.1 Changing axis-1 and 2 working envelope

The robot is equipped with the mechanical stoppers to restrict the working envelop of each axis mechanically. Changing the mechanical working envelope of the robot by changing this mechanical stoppers refers to “working envelope change.”

This section explains how to change axis-1 and 2 working envelope of the robot.

NOTICE

- Never reuse a mechanical stopper that has collided with the arm and was used to change the working envelope. A damaged mechanical stopper will not function properly.
- When designing, manufacturing, and mounting the mechanical stoppers, be sure to observe the following points.
  - Be sure to set the changed working envelope to be within the range shown in the specification table and external view drawing.
    The robot cannot be operated in a working envelope that is outside the specifications.
  - Set the origin position information again by following the procedure in “STEA1211 THE600-E robot manual 14.8 ③ Other Origin Position Settings.”
    If a mechanical stopper is changed, the function described in “STEA1211 THE600-E robot manual 14.8.1.4 Restoring Data by SYSTEM1 or SYSTEM2” is disabled. At installation, enter the original position data by following the procedure in “STEA1211 THE600-E robot manual 14.8.1.2 Setting HOME1 and HOME2.”
  - A function is also available to restore the origin position using a setting jig provided by the customer.
    If using the procedure in “STEA1211 THE600-E robot manual 14.8.1.3 Restoring Data by HOME1 or HOME2” enter the origin position data using the procedure in STEA1211 THE600-E robot manual 14.8.1.2 Setting HOME1 and HOME2” during the installation process.
  - A robot failure caused by a mechanical stopper is outside the scope of the warranty.
- Be sure to set the software limits inside the mechanical stoppers.
  If the software limits are not set inside the mechanical stoppers, the robot will collide with the mechanical stoppers during operation.
- If the robot collides with a mechanical stopper and operation is stopped, immediately turn off the controller power, and contact the SHIBAURA MACHINE Service Department.

The axis-1 working envelope can be changed by changing the position of mechanical stoppers as shown in Fig. 6.5.

Table 6.4 Axis-1 working envelope before and after change

		Before change	After change
Axis-1 working envelope	＋direction	140°	110°
	－direction	140°	110°

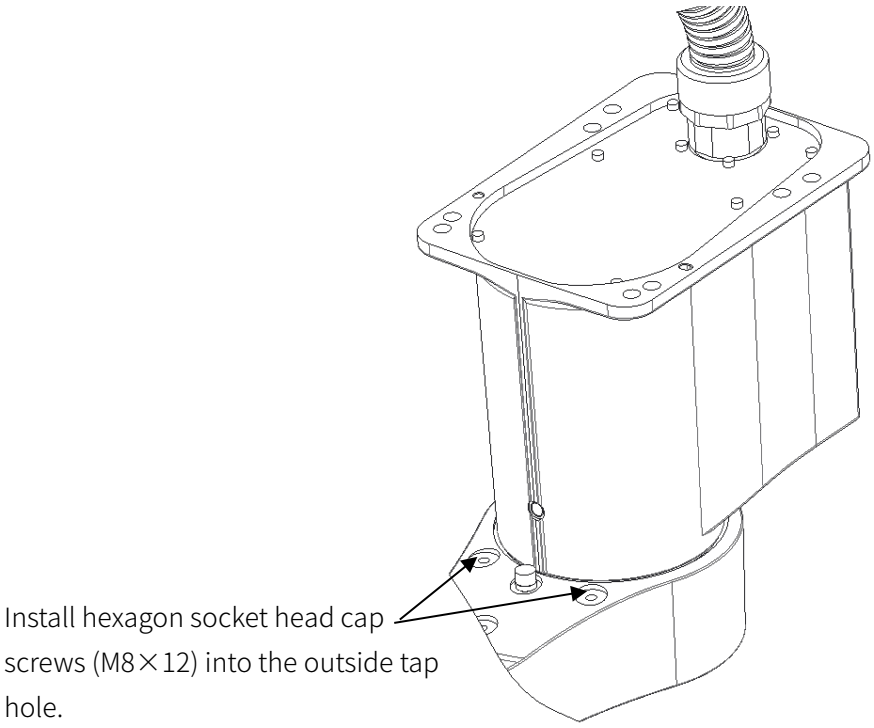


Fig. 6.5 Example of changing axis-1 working envelope

When the axis-2 working envelope is changed, it can be set to any working envelope by changing the shapes of mechanical stoppers. Fig. 6.6 shows the mounting position of the axis-2 mechanical stopper and Fig. 6.7 and Fig. 6.8 show the examples of changing the axis-2 mechanical stopper. The customer should design and manufacture the mechanical stoppers as required.

Table 6.5 Axis-2 working envelope before and after change

		Before change	After change (Example when changed)
Axis-2 working envelope	＋direction	150°	120°
	－direction	150°	150°

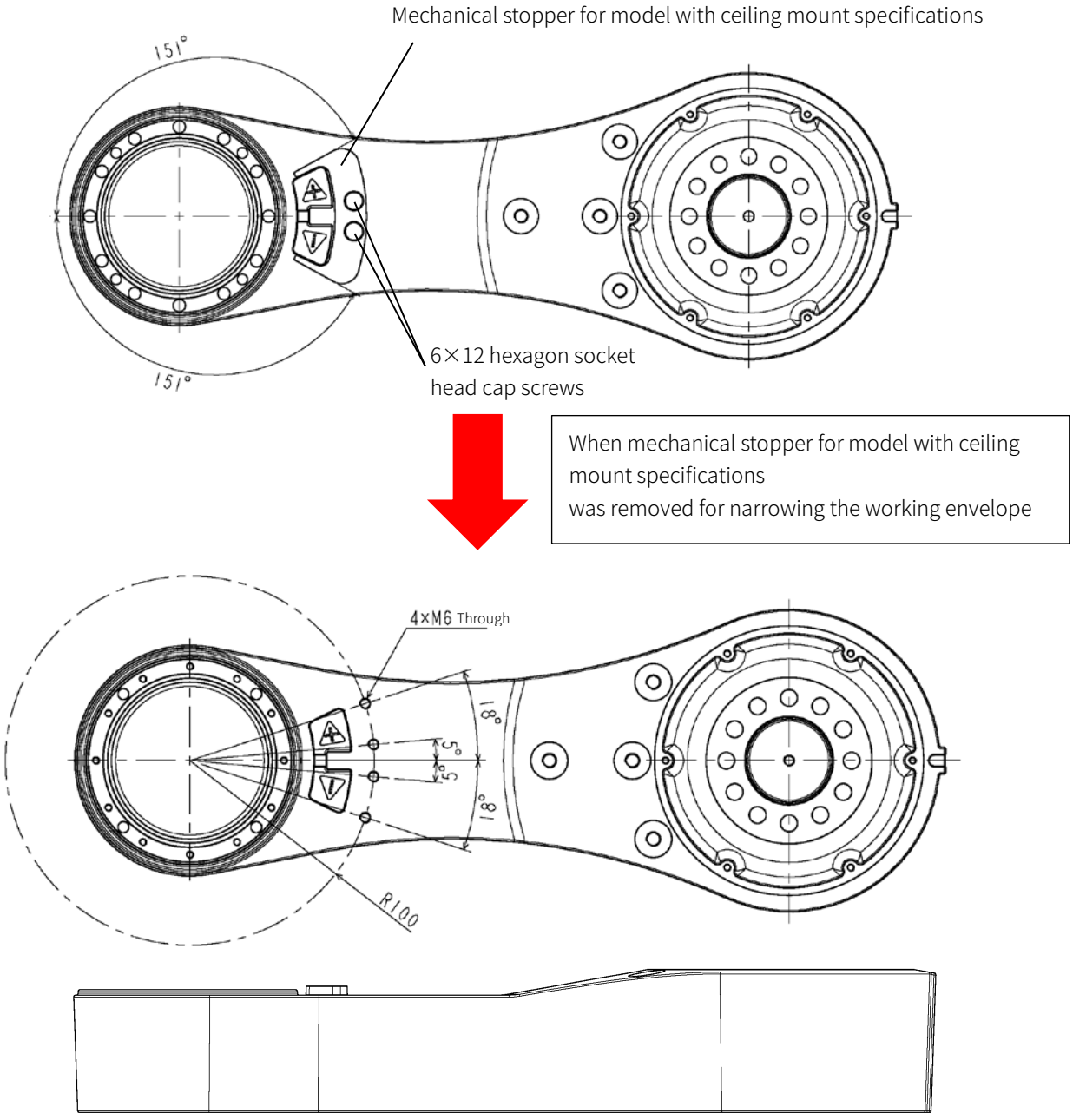


Fig. 6.6 Mounting position of axis-2 mechanical stopper

Example of changing axis-2 mechanical stopper
Set working envelope: 120°, -150°

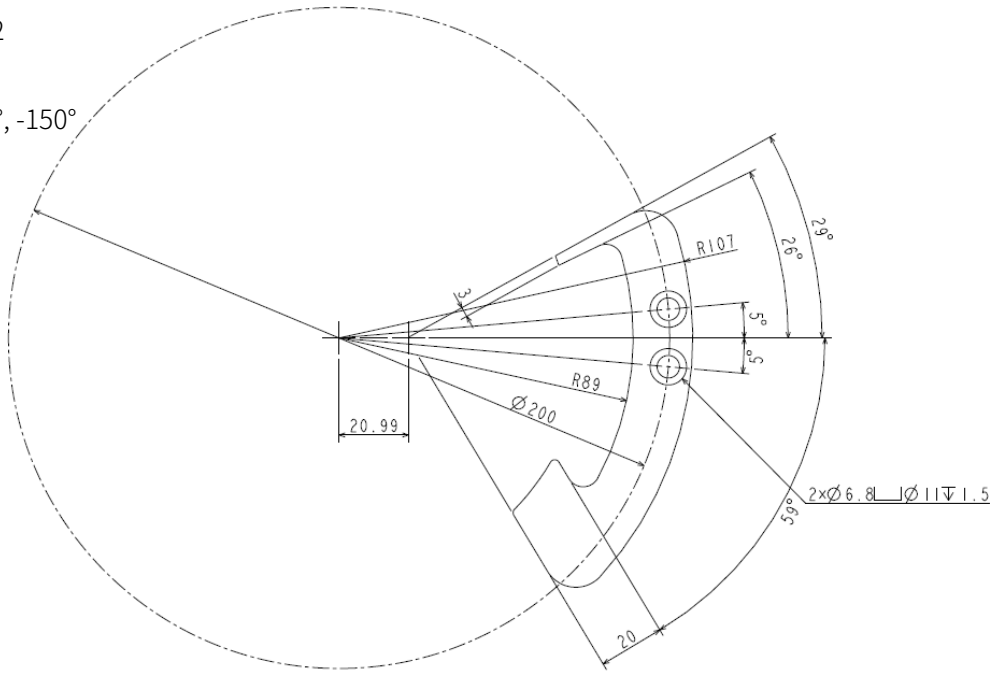


Fig. 6.7 Example of changing axis-2 mechanical stopper

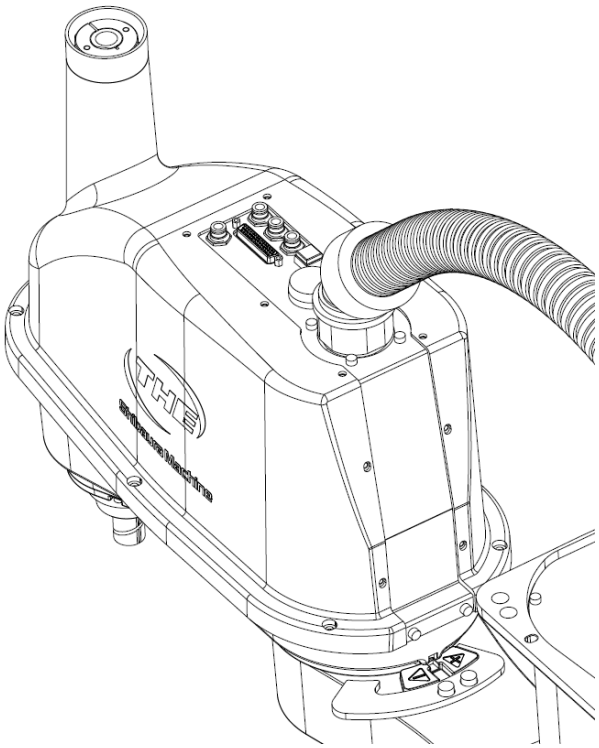


Fig. 6.8 Example of changing axis-2 working envelope

If a working envelope is changed, user parameter must be changed.
For how to change the software limit, see “3.3.3 Changing software limit” in the “STEA1211: THE600-E Robot Manual”.

7. Setting Hand

The worker should receive safety training for laborer engaged in industrial robot-related work defined by each country's laws and regulations.
Be aware that failure and accident resulting from the work done by the customer will not be under our warranty.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	

MEMO

Before starting work, check the information in STEA1211 THE600-E robot manual 10.1 Rescuing a Person Trapped by a Robot, and conduct a rescue drill.

⚠ CAUTION

- Never touch fluid that has leaked from the position detector backup batteries.
  If any fluid gets on your skin or in your eyes, inflammation will occur.
  - Do not bring the robot into high-temperature environments because fluid leakage is more likely to occur.
  - Wear protective gloves if there is a possibility of coming into contact with leaked fluid.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Replace any parts that the fluid has splashed on.

7.1 Mounting hand

The hand mounting specifications for the THE600-T-E are identical to the THE600-E.

For details, see “4.1 User Wiring” in the “STEA1211: THE600-E Robot Manual”.

7.2 Mounting camera and air valve

The camera and air valve mounting specifications for the THE600-T-E are identical to the THE600-E.

For details, see “4.2 User Wiring” in the “STEA1211: THE600-E Robot Manual”.

7.3 Hand air piping

Four hand piping lines (Φ6×4) are available. The lines are connected on the base rear and the arm-2 top.

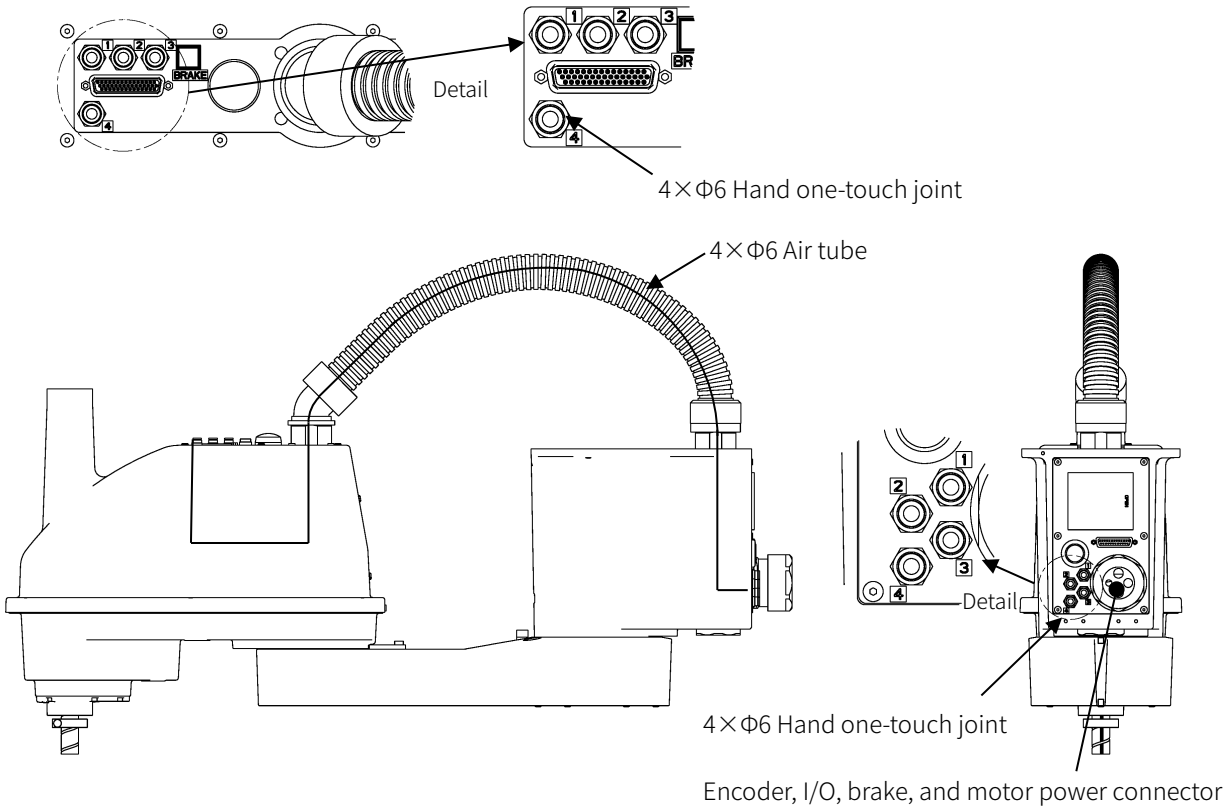


Fig. 7.1 Hand air piping and wiring

MEMO

Numbers are assigned to the joints mounted on the robot.

When mounting the hand air piping, it is recommended that piping numbers be affixed to the mating air tubes to prevent piping mounting mistakes when piping is removed during maintenance work.

NOTICE

- Do not route piping together with the body harness.
  Otherwise, excessive force could be applied on the body harness, resulting in breakage of the internal cable.
- The solenoid valve air tubes are not supplied with the robot.
  They should be provided by the customer.

7.4 Permissible load connections and program setting

The permissible load conditions and program setting specifications for the THE600-T-E are identical to the THE600-E..

For details, see “4.4. Permissible Load Conditions and Program Settings” in the “STEA1211: THE600-E Robot Manual”.

8. User Wiring

The work should receive safety training defined by each country's laws and regulations.
Be aware that the failure and accident resulting from the work done by the customer will not be under our warranty.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	

MEMO

- *Before starting work, check STEA1211: THE600-E Robot Manual 10.1 Rescuing a Person Trapped by a Robot, and conduct a rescue drill.*
- *The 1st, 2nd, and 4th axes of THE600-E can be moved by turning off the drive power (servo off) and pushing them by hand. By releasing the brake while the 3rd axis is powered on, it can be pushed and moved by hand without turning on the drive power supply (turning on the servo). For details, see "9. Teaching."*
- *We are preparing various methods for guiding the robot, which is done by turning on a drive power supply (turning on the servo). For details, refer to the following the separate manuals.*
  - TP5000 Operation Manual
  - TCmini (Internal PLC) Manual
  - Communication Manual
  - TSAssist User's Manual

⚠ CAUTION

- Never touch fluid that has leaked from the position detector backup batteries.
  If any fluid gets on your skin or in your eyes, inflammation will occur.
  - Do not bring the robot into high-temperature environments because fluid leakage is more likely to occur.
  - Wear protective gloves if there is a possibility of coming into contact with leaked fluid.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Replace any parts that the fluid has splashed on.

8.1 Hand wiring

The hand wiring specifications for the THE600-T-E are identical to the THE600-E.
For details, see “5.1 Hand Wiring” in the “STEA1211: THE600-E Robot Manual”.

⚠ DANGER

- When fabricating the hand wiring cable, be sure to use the compatible wire.
  Otherwise, the cable could overheat or short circuit between the terminals, resulting in a fire.
- Never connect a power supply with a voltage exceeding the specified range to the user wiring connector.
  This could cause the cable to overheat, resulting in a fire.
  - Before connecting, use the tester or similar device to check the voltage of the power supply.

NOTICE

- Never mistake the terminal arrangement for the hand wiring.
  Otherwise, the connected parts could fail.
  - Before usage, use the tester or similar device to check the terminal arrangement.
- When fabricating the hand wiring cable, be sure to use the compatible wire.
  Otherwise, overheat or short circuit could occur, resulting in burning of the cable.
- Be sure to use a highly flexible and twisting-resistant robot cable for the hand wiring cable.
  If a general-purpose cable is used, the wire may break during robot operation.
  - Secure the cable to the arm using a saddle or other mechanism to prevent the cable from being swung around.
- Never connect the hand wiring cable to a power supply whose voltage exceeds the specified range.
  Otherwise, the cable could overheat, resulting in a fire.
  - Before connecting, use the tester or similar device to check the voltage of the power supply.
- Be sure to perform hand wiring and piping while checking the following status during robot operation.
  Otherwise, the robot and hand will become faulty early.
  - Check whether a load is being applied to the hand connector.
  - Check whether the hand wiring and piping are contacting the robot or peripheral equipment and are worn.

8.1.1 Hand control wiring

The hand control wiring specifications for the THE600-T-E are identical to the THE600-E.
For details, see “5.1.1 Hand Control Wiring” in the “STEA1211: THE600-E Robot Manual”.

8.1.2 User wiring

Sixteen (16) signals are provided as user wiring that is controlled by electric gripper, separately placed PLC, etc.
The wiring is connected to the connector on the arm-2 top and on the base rear.
The user should prepare the connector and connect the cable.

Connector (standard) type:	<Hood>	XM2S-2511 (manufacturer: OMRON)
	<Connector>	XM3A-2521 (manufacturer: OMRON)

Applicable cross sectional area of wire conductor: AWG 24 to 26 (plain braided wire and twisted wire)
*Connect the connector and wire by soldering.

For the arm-2 side connector, see the connector in “8.1.1 Hand control wiring” in the “STEA1211: THE600-E Robot Manual”.

The signal line from the base rear connector “USER” passes inside the robot and is wired to the hand wiring connector “HAND.”
Connect the manufactured cable connector to “USER” and “HAND.”

MEMO

Hand I/O cable to be connected to the connector “HAND” is also optionally available. For optional hand I/O cable, see “5.2 Hand I/O cable (option)” in the “STEA1211: THE600-E Robot Manual”.

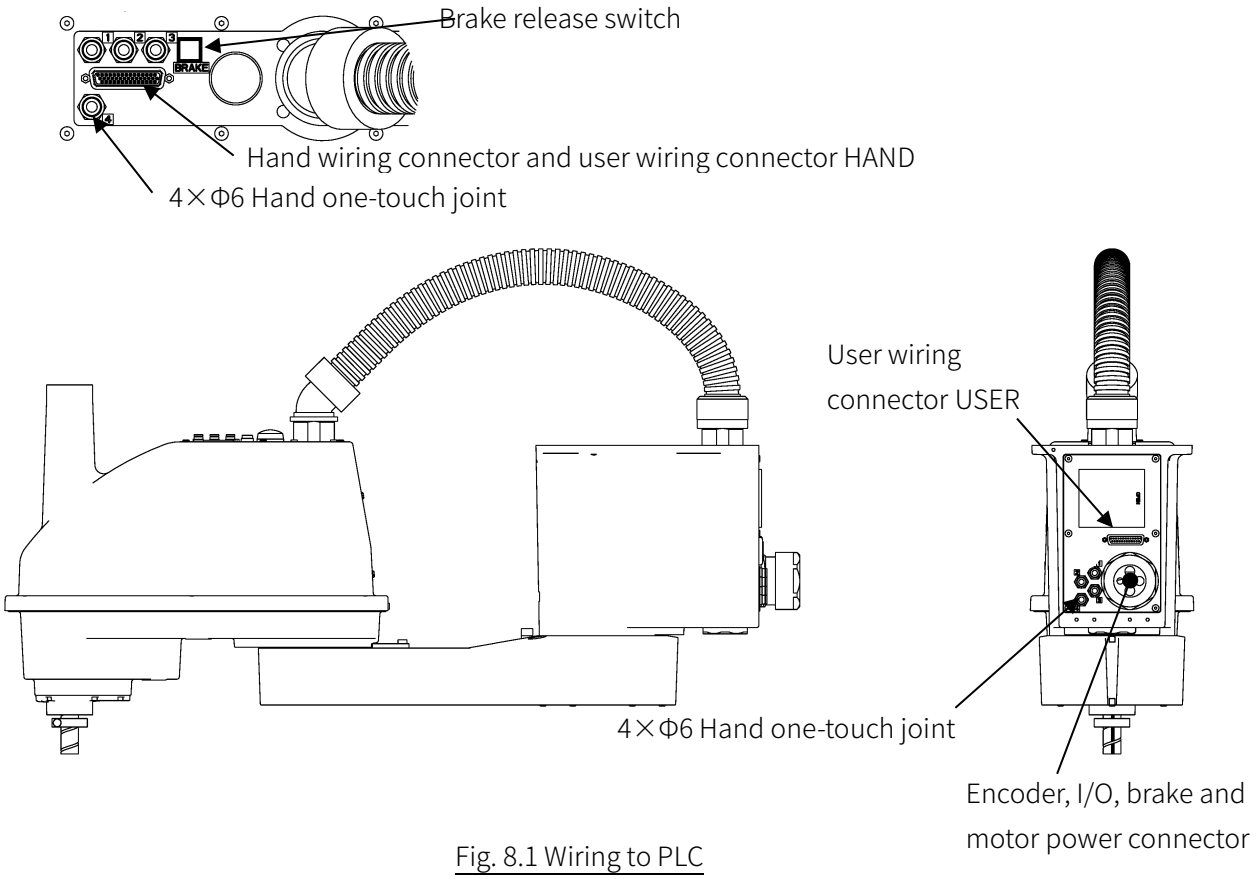


Fig. 8.1 Wiring to PLC

User wiring is as shown in Fig. 8.2.

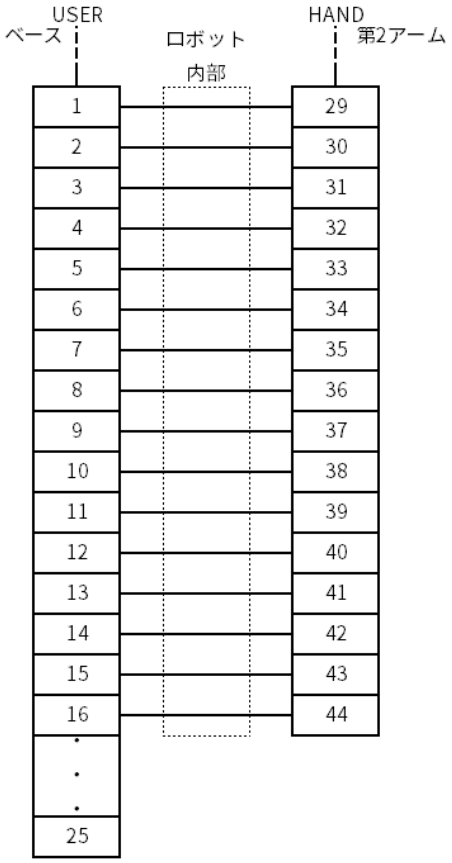
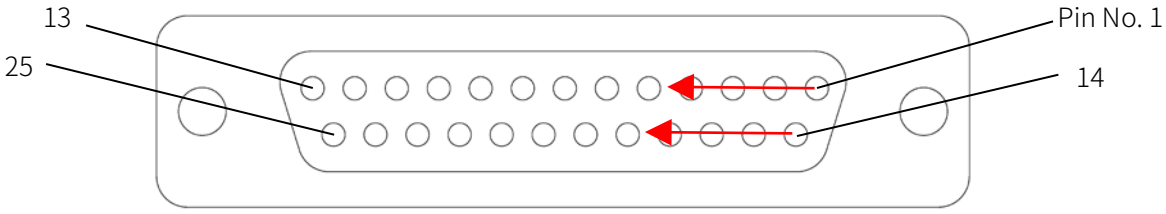


Fig. 8.2 User wiring diagram



Pin arrangement as viewed from the connection port

Fig. 8.3 User connector pin numbers

MEMO

For the pin arrangement on the HAND connector side, see STEA1211: THE600-E Robot Manual Fig. 8.5.

8.1.3 Specifications of hand connector HAND

The hand connector specifications for the THE600-T-E are identical to the THE600-E..
For details, see “8.1.3. Hand Connector Specifications” in the ““STEA1211: THE600-E Robot Manual”.

8.1.4 Connection example

The hand wiring specifications for the THE600-T-E are identical to the THE600-E.
For details, see “8.1.4 Connection Examples” in the “STEA1211: THE600-E Robot Manual.

9. Teaching

Teaching for the THE600-T-E are identical to the THE600-E..
For details, see “9.Teaching” in the ““STEA1211: THE600-E Robot Manual”.

10. Operation

Operation for the THE600-T-E are identical to the THE600-E..
For details, see “10. Operation” in the ““STEA1211: THE600-E Robot Manual”.

11. Repacking

Repacking for the THE600-T-E are identical to the THE600-E..
For details, see “11. Repacking” in the ““STEA1211: THE600-E Robot Manual”.

Maintenance Manual

This manual describes how to maintain respective units of the robot.

12. Maintenance

The arm mounting direction is different for models with ceiling mount specifications, but the basic structure is identical to the THE600-E..

The items that differ from the THE600-E. are described below.

For details about the other inspection items, see “MAINTENANCE” in the “STEA1211: THE600-E Robot Manual”.

13. Maintenance of the Robot Body

The worker should receive safety training for laborer engaged in industrial robot-related work defined by each country's laws and regulations. SHIBAURA MACHINE Service Department also accepts replacement work at an extra cost.

Be aware that failure and accident resulting from the work done by the customer will not be under our warranty.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	

MEMO

Before starting work, check the information in STEA1211: THE600-E Robot Manual 10.1 Rescuing a Person Trapped by a Robot, and conduct training.

⚠DANGER

- Water and other liquids must not be brought into the area around the robot.
  If a liquid is spilled, it could result in an electric shock.
- Do not put devices that generate excessive amounts of noise near the robot.
  Otherwise, a robot could malfunction due to noise and collide with a worker.
- Before turning on the servo for the robot, always make sure that there is no one inside the robot's safety protection zone.
  Otherwise, a collision could occur with a person due to a movement by the robot.
  - When going inside, leave the safety door open, carry a teach pendant set to MANUAL mode, and post that work is currently in progress.
  - You can check the status of the servo using the LED lamp on the robot.

- Be sure to perform maintenance and inspection periodically.
  Failure to perform maintenance and inspection could lead to an unpredictable accident due to a robot malfunction.

⚠ WARNING

- When performing an operation to initialize all output signals (such as the system input signal “Output signal initialization” or RESET DOUT command), be sure to connect the hand to the hand output signals (DOUT201 to DOUT208).
  Otherwise, the robot hand will move.
  - The status of the hand output signals does not change even if the output signals are all initialized.
  - Be sure to use a robot hand having a structure and wired so that it will not release the workpiece or other object even if the controller power or hand driving power supply is cut off.
  - Conduct daily inspection on the air tube connected to the hand, and replace it if it shows any damage.
  - Install a fence or other protection for prevent dropped and flying objects.
- Before performing work where the robot is touched, be sure to always check that the surface temperature is less than 50°C.
  Otherwise, a burn could occur due to touching a hot motor or the area around the motor.

⚠ CAUTION

- Never touch fluid that has leaked from the position detector backup batteries.
  If any fluid gets on your skin or in your eyes, inflammation will occur.
  - Do not bring the robot into high-temperature environments because fluid leakage is more likely to occur.
  - Wear protective gloves if there is a possibility of coming into contact with leaked fluid.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Replace any parts that the fluid has splashed on.

13.1 Details of Inspection

13.1.1 Check of Each Bolt (or Screw) for Clamping

13.1.1.1 Check of Hand Set bolts

Check of Hand Set bolts for the THE600-T-E are identical to the THE600-E .

For details, see “13.1.1.1 Check of Hand Set bolts in the ““STEA1211: THE600-E Robot Manual ”.

13.1.1.2 Robot installation bolts

Robot installation bolts for the THE600-T-E are identical to the THE600-E .

For details, see “13.1.1.2 Robot installation bolts in the ““STEA1211: THE600-E Robot Manual ”.

13.1.1.3 カバー取付ボルトの確認

Check the looseness of the bolts for mounting the covers of the robot.

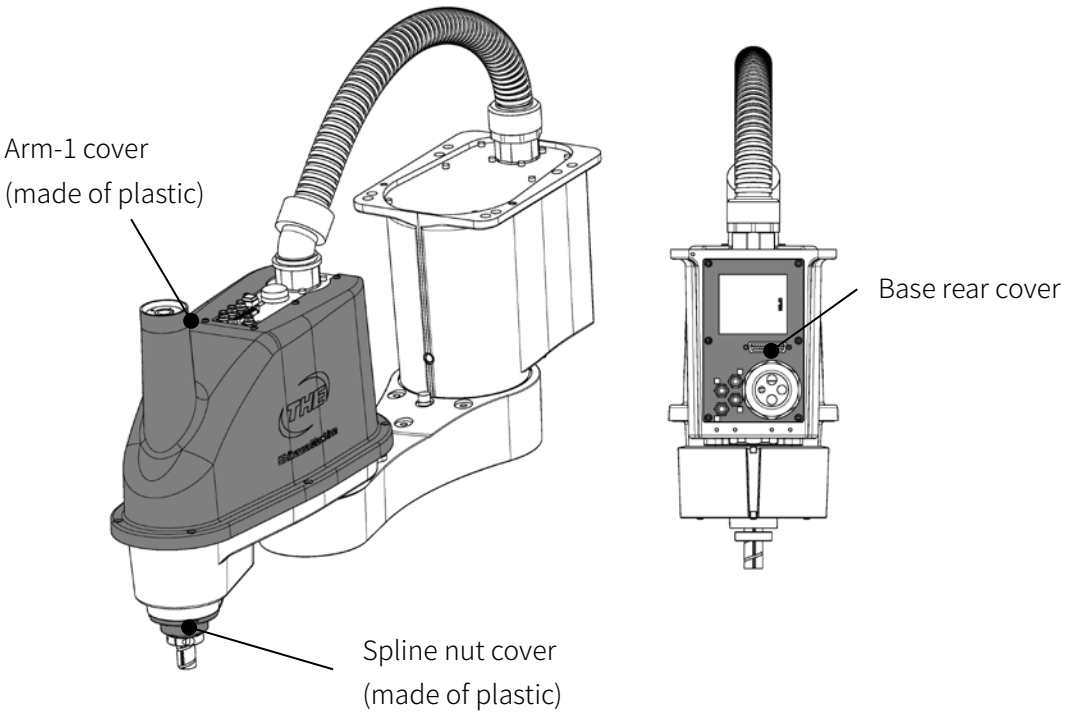


Fig. 13.1 Robot covers

MEMO

When tightening the bolts for attaching the plastic robot covers, if the tightening torque is too high, it could cause the robot cover to crack. Tighten the bolts by referring to the tightening torques shown in the diagrams indicating the mounting bolts for the various covers.

13.1.2 Check of Duct Connector and Cable Clamps for Clamping

Check of Duct Connector and Cable Clamps for Clamping for the THE600-T-E are identical to the THE600-E.

For details, see “13.1.2 Check of Duct Connector and Cable Clamps for Clamping in the ““STEA1211: THE600-E Robot Manual”.

13.1.3 Check of Operation of Each Axis

Check of Operation of Each Axis for the THE600-T-E are identical to the THE600-E.

For details, see “13.1.3 Check of Operation of Each Axis in the ““STEA1211: THE600-E Robot Manual”.

13.2 Maintenance of the Ball Screw Spline Unit

Maintenance of the Ball Screw Spline Unit for the THE600-T-E are identical to the THE600-E.

For details, see “13.2 Maintenance of the Ball Screw Spline Unit in the ““STEA1211: THE600-E Robot Manual”.

14. Robot Zero Point and Position Detector Error

The worker should receive safety training for laborer engaged in industrial robot-related work defined by each country's laws and regulations.
Be aware that SHIBAURA MACHINE's warranty does not cover failures and accidents resulting from the work done by the customer.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object. Protect from collision with the arm.	
Safety glasses	Protection part: Eyes Use: Protect from a flying object. Protect from collision with the arm.	
Protective gloves	Protection part: Hands and fingers Use: Protect hands and fingers when caught in the machine. Prevent a carried object from dropping.	
Protective shoes	Protection part: Feet and toes Use: Protect from a falling object.	

MEMO

Before starting work, check STEA1211: THE600-E Robot Manual 10.1 Rescuing a Person Trapped by a Robot, and conduct training.

⚠ DANGER

- Water and other liquids must not be brought into the area around the robot.
  If a liquid is spilled, it could result in an electric shock.
- Do not put devices that generate excessive amounts of noise near the robot.
  Otherwise, a robot could malfunction due to noise and collide with a worker.
- Before turning on the servo for the robot, always make sure that there is no one inside the robot's safety protection zone.
  Otherwise, a collision could occur with a person due to a movement by the robot.
  - When going inside, leave the safety door open, carry a teach pendant set to MANUAL mode, and post that work is currently in progress.
  - You can check the status of the servo using the LED lamp on the robot.
- Be sure to perform maintenance and inspection periodically.
  Failure to perform maintenance and inspection could lead to an unpredictable accident due to a robot malfunction.

⚠ WARNING

- When performing an operation to initialize all output signals (such as the system input signal “Output signal initialization” or RESET DOUT command), be sure to connect the hand to the hand output signals (DOUT201 to DOUT208).
  Otherwise, the robot hand will move.
  - The status of the hand output signals does not change even if the output signals are all initialized.
  - Be sure to use a robot hand having a structure and wired so that it will not release the workpiece or other object even if the controller power or hand driving power supply is cut off.
  - Conduct daily inspection on the air tube connected to the hand, and replace it if it shows any damage.
  - Install a fence or other protection for prevent dropped and flying objects.
- Before performing work where the robot is touched, be sure to always check that the surface temperature is less than 50°C.
  Otherwise, a burn could occur due to touching a hot motor or the area around the motor.

⚠ CAUTION

- Never touch fluid that has leaked from the position detector backup batteries.
  If any fluid gets on your skin or in your eyes, inflammation will occur.
  - Do not bring the robot into high-temperature environments because fluid leakage is more likely to occur.
  - Wear protective gloves if there is a possibility of coming into contact with leaked fluid.
  - If the fluid gets on your skin, rinse it off immediately by flushing with water.
  - If the fluid gets in your eyes, rinse it off immediately by flushing with water, and then seek medical attention.
  - Replace any parts that the fluid has splashed on.

14.1 Zero Position

This section describes the zero position required for zero setting. When moving the axes 1, 2, and 4 of the robot to the zero position, see “14.1.1 Locations of Zero Point Match-Marks of the Robot” and "STA1171: THE600 Robot Manual", and align the zero point match-mark of each axis. The axis 3 does not have a zero point match-mark. See “8.5.2 Axis-3 Zero Position” in the “STEA1211: THE600-E Robot Manual” to move it to the zero position. The installation of the zero setting tool (optional) is explained in “14.1.2 Installing the Zero Setting Tool (Optional).”

14.1.1 Locations of Zero Point Match-Marks of the Robot

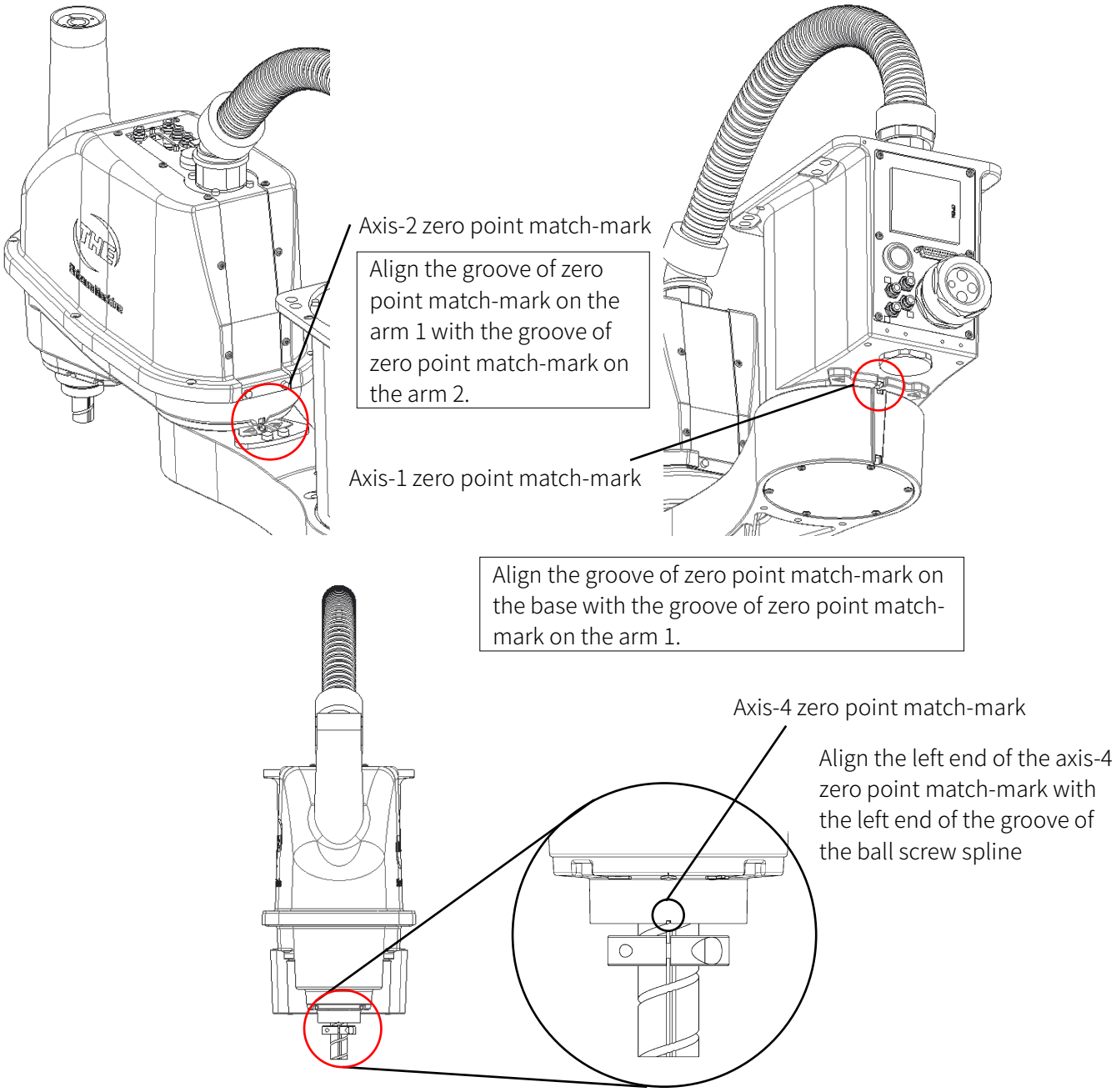


Fig. 14.1 Locations of zero point match-marks

14.1.2 Installing the Zero Setting Tool (Optional)

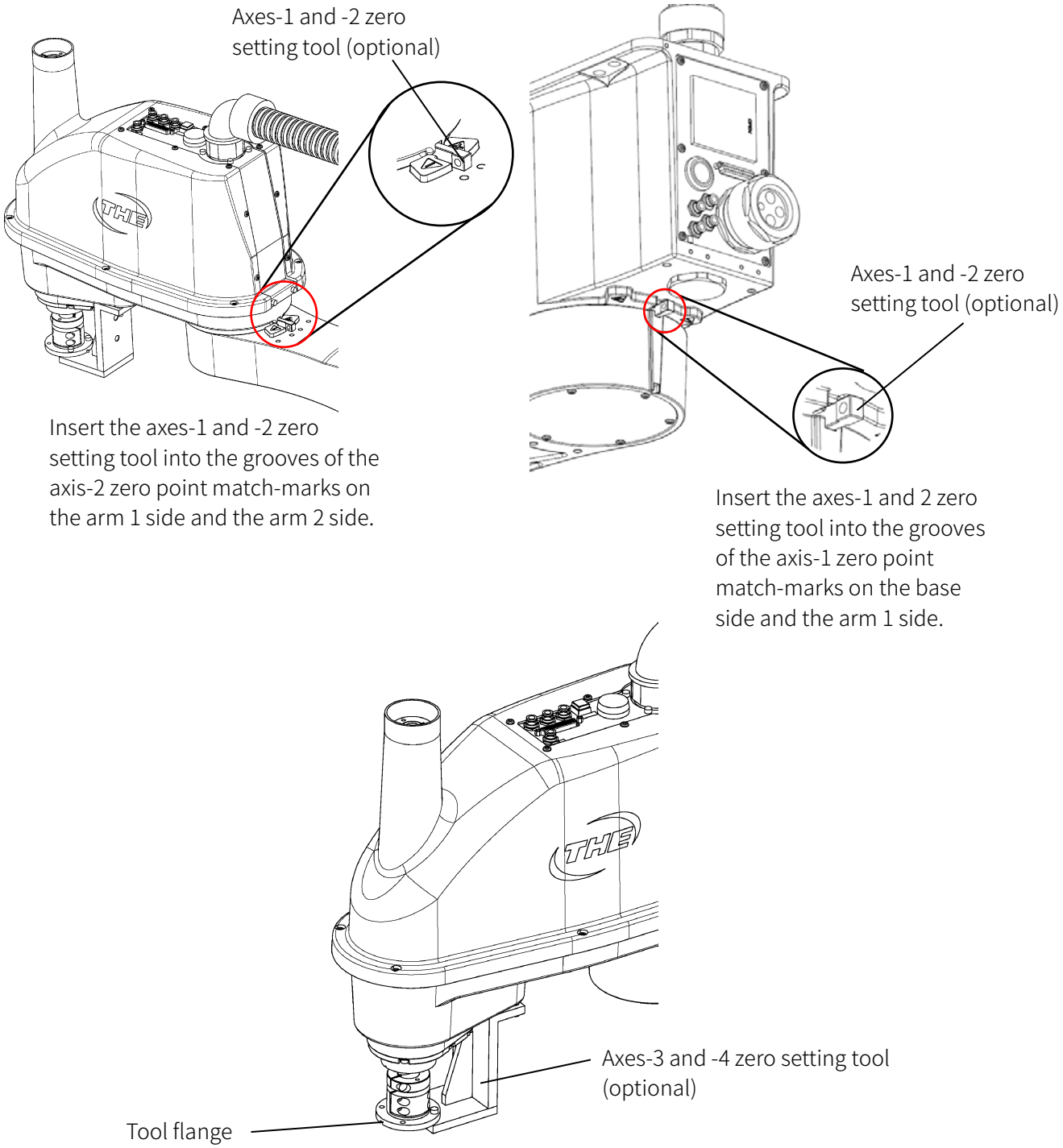


Fig. 14.2 Installing the Zero Setting Tool (Optional)

15. Replacement Parts for Maintenance

15.1 Notes on Replacement Parts for Maintenance

MEMO

Before starting work, check the information in STEA1211: THE600-E Robot Manual 10.1 Rescuing a Person Trapped by a Robot, and conduct training.

⚠DANGER

- Be sure to replace parts with the SHIBAURA MACHINE-specified service replacement parts.
  Use of unauthorized parts could lead to an unpredictable accident.
  - All items other than the position detector batteries are special-specification products. To order them, please contact SHIBAURA MACHINE.
  - Be sure to contact the SHIBAURA MACHINE service company to request any replacement work other than replacement of the position detector batteries. A fee will be charged for any replacement work.
  - SHIBAURA MACHINE does not cover any accidents and damage resulting from replacement work performed by the customer and its results.

NOTICE

- Be sure to replace parts with the SHIBAURA MACHINE-specified maintenance replacement parts.
  Use of unauthorized parts could lead to reduced robot performance and lifespan. The robot will malfunction.

15.2 List of Replacement Parts – Robot

The service parts list is shown below.

No	Part name	Model	Drawing No.	Unit code	Manufacturer	Q'ty	Remarks
1	AC servo motor		S968416	Y610D0CW0	SHIBAURA MACHINE	1	Axis 1
2			S968417	Y610D0CX0		1	Axis 2 and Axis 4
3			S968418	Y610D0CY0		1	Axis 3
5	Reduction gear		S968522	Y610D0CZ0	SHIBAURA MACHINE	1	Axis 1
6			S981478	Y610D0D00		1	Axis 2
8	Timing pulley		S968379	Y610D0DD0	SHIBAURA MACHINE	1	Axis 3 (ball screw side)
9			S968380			1	Axis 3 (motor side)
10			S968381			1	Axis 4 (lower side of intermediate shaft)
11			M342404			1	Axis 4 (ball screw side)
12			M342405			1	Axis 4 (motor side)
13			M342406			1	Axis 4 (upper side of intermediate shaft)
14	Timing belt		S968393			1	Axis 3
15			S968395			1	Axis 4 (ball screw side)
16			S968397			1	Axis 4 (motor side)
17	Ball screw spline unit		H880457	Y610D0CP0	SHIBAURA MACHINE	1	
18	Body harness		F127810	Y610D0DF0	SHIBAURA MACHINE	1	
19	LED Lamp		M342499	Y610D0DU0	SHIBAURA MACHINE	1	
20	O ring	CO0540A			NOK Corp.	1	Axis 2
21		CO0543A				1	Axis 2
22		CO0548A				1	Axis 1

No	Part name	Model	Drawing No.	Unit code	Manufacturer	Q'ty	Remarks
23	Grease	Multemp FZNo.00			Kyodo Yushi Co., Ltd.		Axis 1
24		SFB No.1 (For reduction gear)			Nidec-Shimpo Corp.		Axis 2
25		AFF grease (For ball screw)			THK		Axis 3
26		LGAF 3E (For key)			Japan SKF Co., Ltd.		All axes
27	Encoder backup battery	Alkaline AA-cell battery				3	All axes
28	Zero setting tool*1		M342399	Y610D0DQ0	SHIBAURA MACHINE		Axis 1 and Axis 2
29			L23752G01	Y610D0DR0			Axis 3 and Axis 4
30	Tension adjustment tool*2		L27611G01	Y610D0DW0	SHIBAURA MACHINE		Axis 4
31			L27612G01	Y610D0DX0			Axis 3 and Axis 4
32	Tool flange		L12515G01	Y610D0300	SHIBAURA MACHINE		

16. Disposal

Disposal for the THE600-T-E are identical to the THE600-E.

For details, see "16. Disposal in the ""STEA1211: THE600-E Robot Manual".